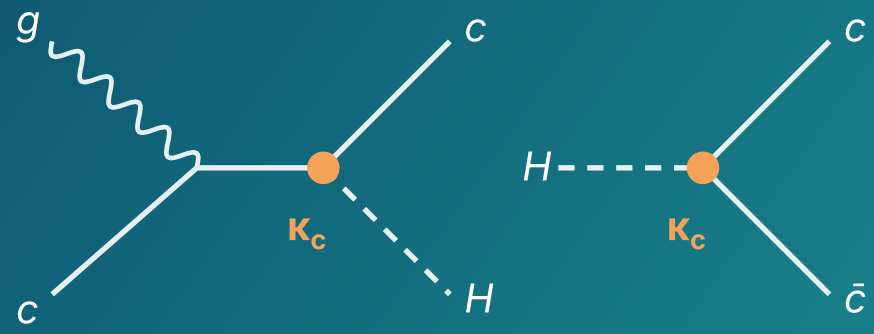
Constraints on Higgs–charm couplings at CMS

Higgs 2026 · Louvain-la-Neuve · 28 October 2026

Chirayu Gupta
Vrije Universiteit Brussel

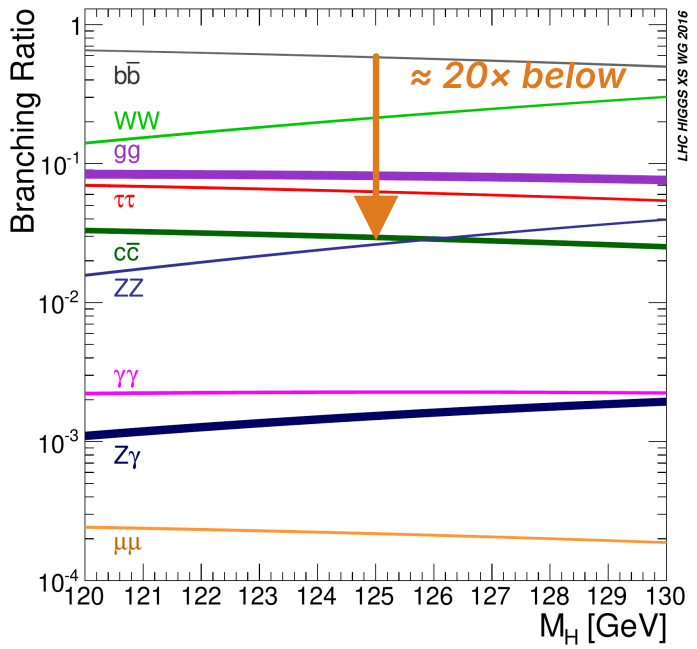
on behalf of the CMS Collaboration



Why the Higgs–charm coupling?

Second-generation quark Yukawas are the open frontier

- Since 2012: couplings to **t, b, τ** measured, all SM-like
- Evidence for **H → μμ** (backup)
- **Charm is the most accessible second-generation quark**



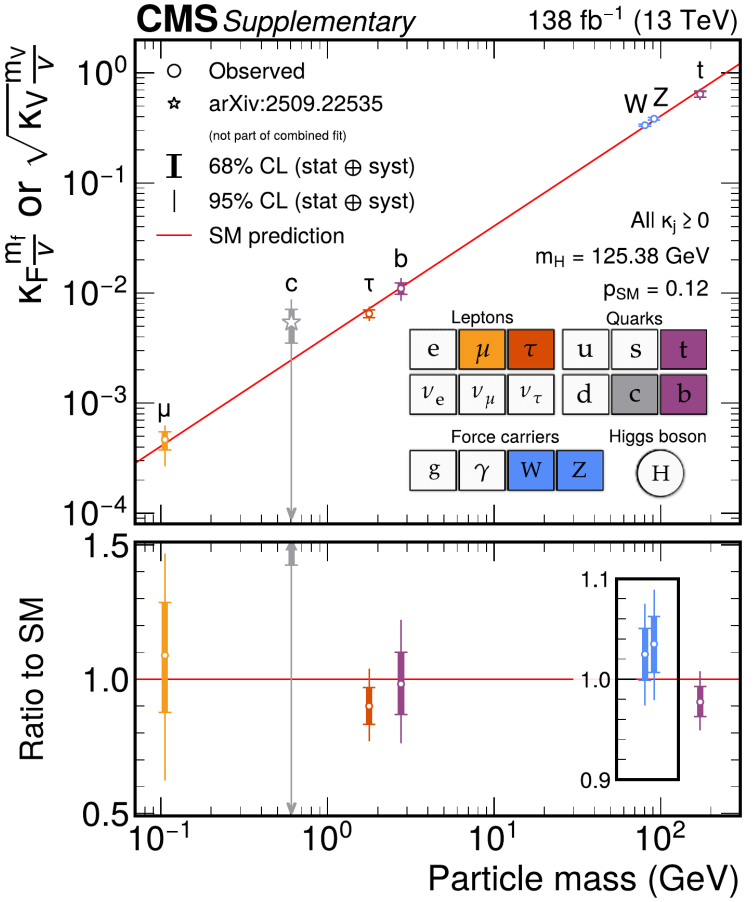
$\mathcal{B}(H \to c\bar{c}) \approx$ **2.9%**, about **20× below** $b\bar{b}$, with huge QCD backgrounds and hard c tagging (slide 3)

$$\kappa_c = y_c / y_c^{SM} \quad \text{(SM: 1)}$$

Flavour-compatible BSM, e.g. spontaneous flavour violation, allows $|\kappa_c| \sim 3$: **current sensitivity is entering this region**

SFV: PRL 123 (2019) 031802
κ_c in SFV: JHEP 02 (2025) 067

LHC Higgs XS WG, Yellow Report 4, arXiv:1610.07922



CMS, Rep. Prog. Phys. 89 (2026) 087801 (suppl.); κ_c point: separate $t\bar{t}H$+VH result, not in the fit

➜ Does the Higgs give charm its mass?

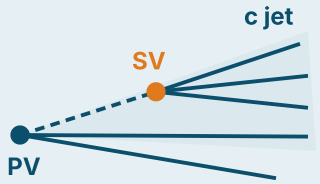
CMS-DP-2025-081

Charm tagging

Finding c jets between b and light jets

Charm signature

c-hadron mass and lifetime lie **between light and b**: a secondary vertex (SV) like b jets, but closer to the primary vertex, so **easily mistaken for light jets**



Algorithm evolution at CMS

Likelihood / BDT

JP
likelihood of track impact parameters

CSV, CSVv2
likelihood ratio, then MLP

Run 2 c tagger
BDT: CvsL, CvsB

Deep NN

DeepCSV
dense DNN (4 × 100), b/c/light classes

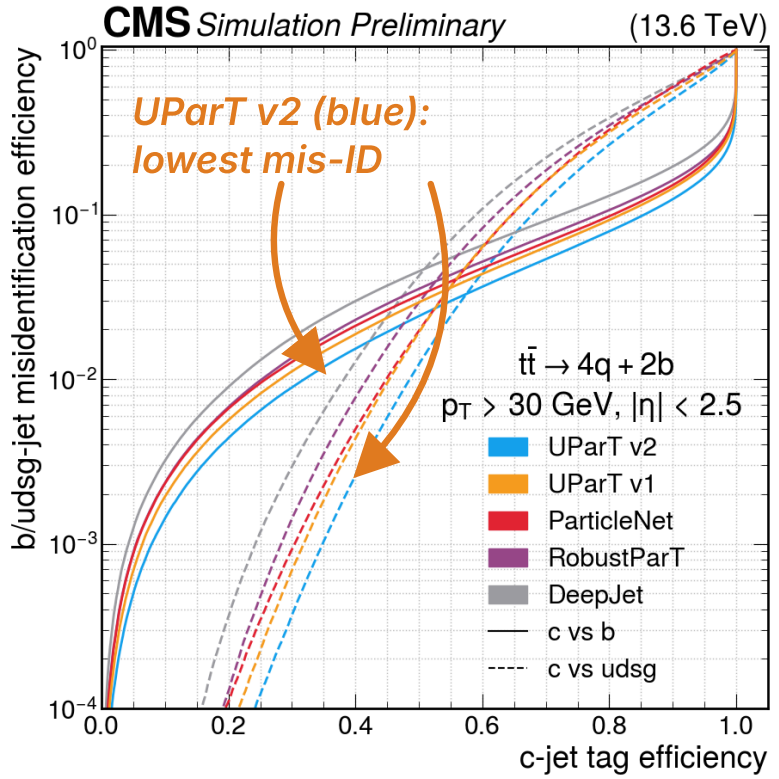
DeepJet
1×1 convolutions + LSTMs on all jet constituents

Particle cloud

ParticleNet
graph NN (dynamic graph CNN)

UParT
transformer with pairwise interaction features, adversarial training

Run 1 → Run 3: likelihoods and trees → dense nets → graphs and attention



$t\bar{t}$ simulation, 13.6 TeV · solid: c vs b, dashed: c vs light

In this talk **DeepJet · VH resolved, cH** **ParticleNet · VH merged, $t\bar{t}$H** **DeepDoubleX · boosted H**

How CMS probes κ_c

In the decay, in production, and through the Higgs width

Direct $H \to c\bar{c}$

slides 5–11

rate $\propto \kappa_c^2/\Gamma_H$ · VH, $t\bar{t}H$, boosted (mostly ggH)

$|\kappa_c| < 3.5$ (2.7) $t\bar{t}H + VH$

PRL 131 (2023) 061801 · PRL 136 (2026) 011801 · PRL 131 (2023) 041801

H + charm production

slides 12–14

cg → cH: κ_c in the production vertex · H → γγ, WW

$|\kappa_c| < 47$ (51)

JHEP 11 (2025) 060 · JHEP 08 (2026) 051

Indirect: Higgs rates and width

slide 15

on-shell and **new** off-shell H → ZZ → 4ℓ

$\kappa_c \in [-3.3, 2.8]$ … $[-8.6, 7.5]$ by assumptions

PRD 112 (2025) 112001 · arXiv:2607.23352

Rare exclusive decays

slide 15

H → J/ψ γ, ψ(2S) γ: sensitive to the *sign* of κ_c

$-166 < \kappa_c < 208$

PLB 865 (2025) 139462

$\mu = \sigma\mathscr{B}/(\sigma\mathscr{B})_{SM}$ · Γ_H = total Higgs width · μ limits and κ_c intervals at 95% CL, observed (expected)

→ All κ_c results: Run 2 (13 TeV, up to 138 fb^{-1}) · Run 3 so far: better triggers (slide 17)

VH(H → cc̄): strategy

PRL 131 (2023) 061801 · CMS-HIG-21-008

Cleanest signature: the W or Z decays to leptons

0ℓ · Z → νν **1ℓ · W → ℓν** **2ℓ · Z → ℓℓ**

Main backgrounds V+jets and tt̄, normalised in control regions

below $p_T^H = 300$ GeV *above*

Resolved · two small-R jets (R = 0.4)

- **>95% of the signal**, but more background
- **DeepJet** CvsL and CvsB: the two highest-CvsL jets form the H candidate
- The Higgs mass peak is up to **20% sharper**, thanks to a c-jet energy regression, recovery of FSR jets and, in 2ℓ, a kinematic fit

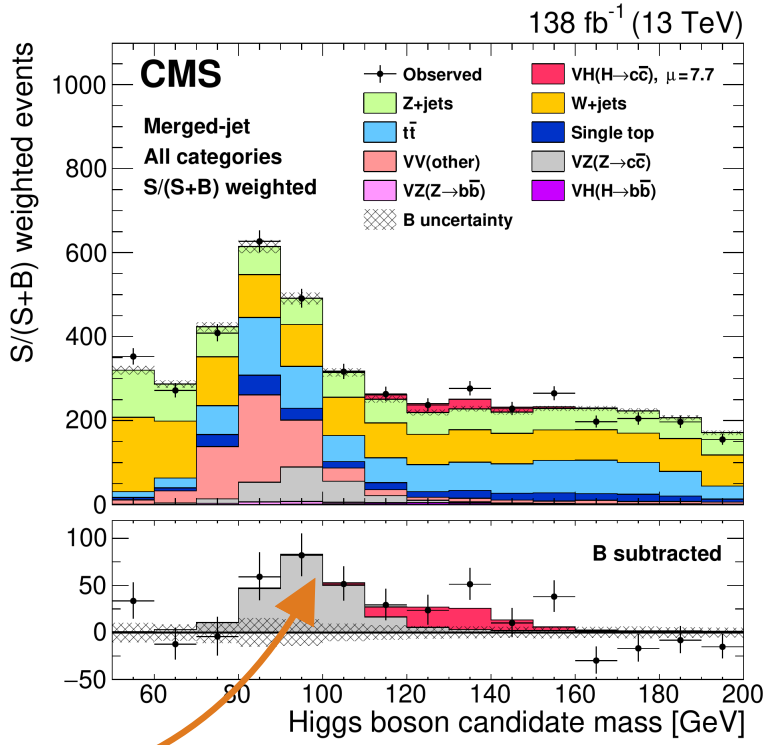
Fit: BDT score (m(H_{cand}) among the inputs)

Merged · one large-R jet (R = 1.5)

- **<5% of the signal**, but much less background
- **ParticleNet** cc̄ tagger (graph NN, mass-decorrelated): **4–7×** better rejection than DeepAK15
- ParticleNet mass regression: **≈50% better** mass resolution

Fit: m(H_{cand}) (BDTs decorrelated from it)

Merged-jet candidate mass; bottom panel after background subtraction: VZ (grey, ≈ 90 GeV) and VH (red)

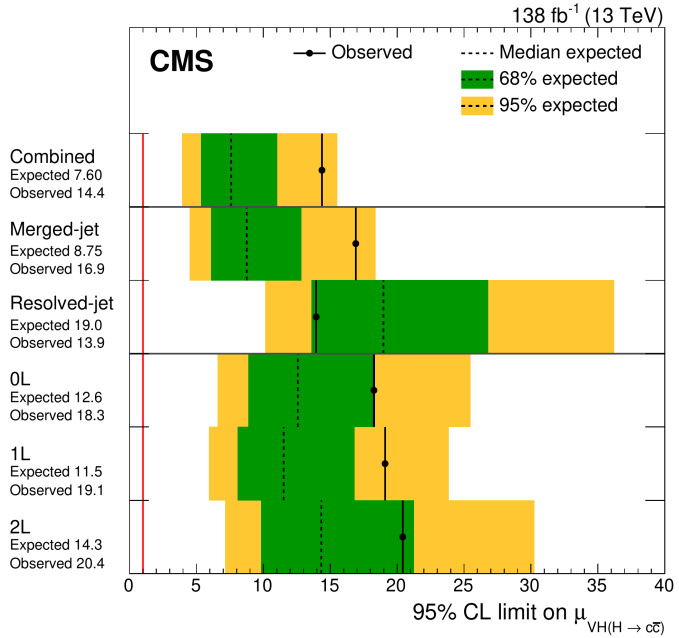


Validation: the same analysis sees **VZ(Z → cc̄) at 5.7σ** (grey peak, right): c tagging and background model work → backup

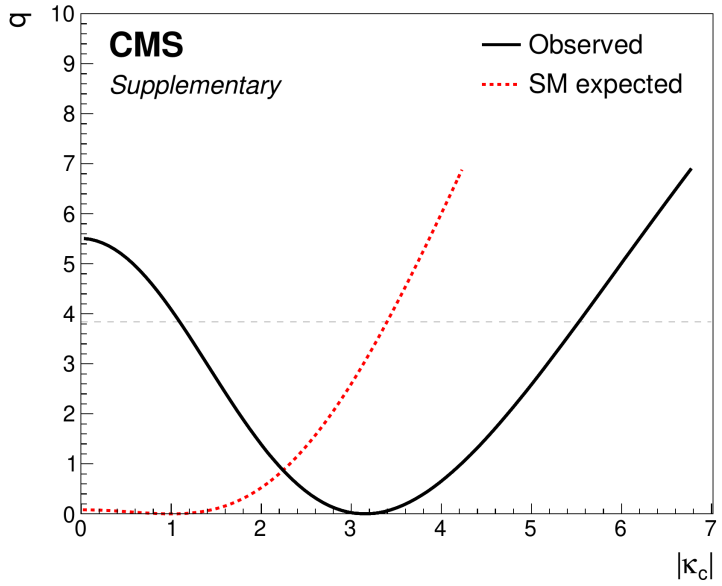
VH(H → $c\bar{c}$): results

PRL 131 (2023) 061801 · CMS-HIG-21-008

The most sensitive single channel of Run 2



95% CL limits on μ per category · red line: SM (μ = 1) · the **merged-jet** category carries most of the sensitivity



q = −2Δln L vs $|\kappa_c|$ (supplementary) · 95% CL where q < 3.84 (dashed line)

85%
of the uncertainty is statistical

Statistics-limited: more data helps directly

Statistical	85%
Experimental	48%
Theory	22%

share of the total uncertainty on μ, in quadrature · details → backup

95% CL · observed (expected)

μ < 14 (7.6) **1.1 < $|\kappa_c|$ < 5.5** (< 3.4)

μ limits with CLs; κ_c interval from the profile-likelihood scan

➜ Expected μ < 7.6: the best single channel of Run 2

Boosted H → $c\bar{c}$

PRL 131 (2023) 041801 · CMS-HIG-21-012

Gluon-fusion enriched: one large-radius jet at high p_T

① Boosted regime: p_T^H > 450 GeV

- High p_T suppresses the QCD background
- H → $c\bar{c}$ in **one large-radius jet** (R = 0.8); soft drop grooms it and sharpens the mass m_{SD}
- Signal: 55% ggF, 25% VBF, 20% VH

② DeepDoubleX tagger

- Convolutional + recurrent network on tracks and secondary vertices
- **Passing** region (signal-rich) and **failing** region (background-rich)
- Mass-decorrelated: QCD in the passing region is **predicted from data** in the failing region

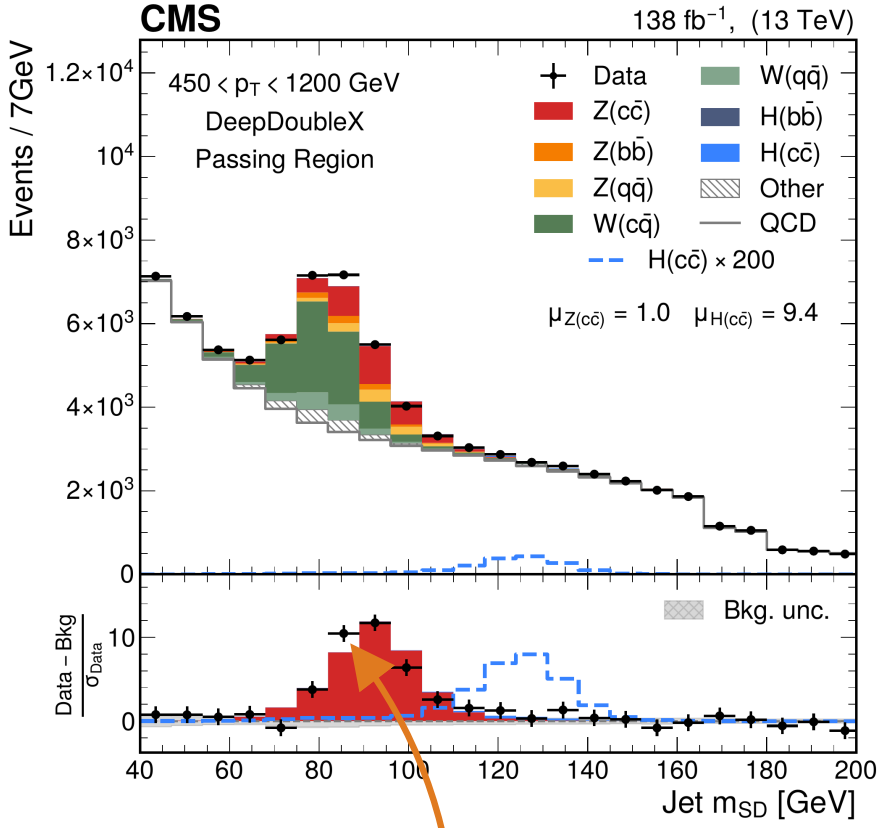
③ Statistical analysis

- Simultaneous binned fit of m_{SD} in both regions, in 6 p_T bins
- Upper limit from the profile likelihood ratio (CLs)

$$\sigma\mathcal{B} / (\sigma\mathcal{B})_{SM} < 47\ (39)$$

95% CL · observed (expected) · no κ_c interpretation in the paper

Jet mass in the passing region · H → $c\bar{c}$ dashed (×200)



Z → $c\bar{c}$ observed for the first time in Z+jets, well over 5σ

$t\bar{t}H(H \to c\bar{c})$: selection and jet tagging

PRL 136 (2026) 011801 · CMS-HIG-24-018

Three $t\bar{t}$ decay channels, every jet flavour-tagged

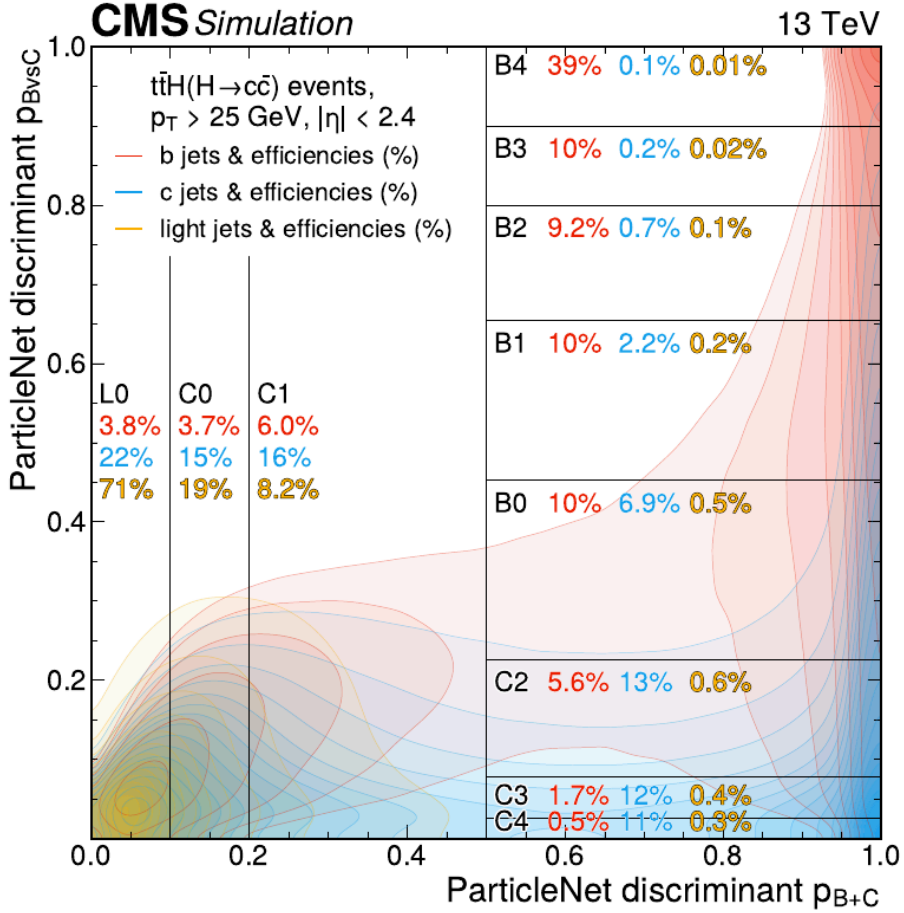
0ℓ — **≥ 7 jets**, H_T > 500 GeV
fully hadronic $t\bar{t}$

1ℓ — **1 lepton + ≥ 5 jets**
single-lepton $t\bar{t}$

2ℓ — **2 opposite-charge leptons + ≥ 4 jets**, Z veto
dilepton $t\bar{t}$

ParticleNet (graph NN) tags every jet as b, c or light: **11 categories** (right), up to **2×** better background rejection than DeepJet

Main background: $t\bar{t}$+jets, split by the flavour of the extra jets: $t\bar{t}$ +b, +≥2b, +c, +≥2c, +light



b, c and light jets in the ParticleNet (b+c vs light, b vs c) plane, with the 11 tagging categories

$t\bar{t}H(H \to c\bar{c})$: event classifier and regions

PRL 136 (2026) 011801 · CMS-HIG-24-018

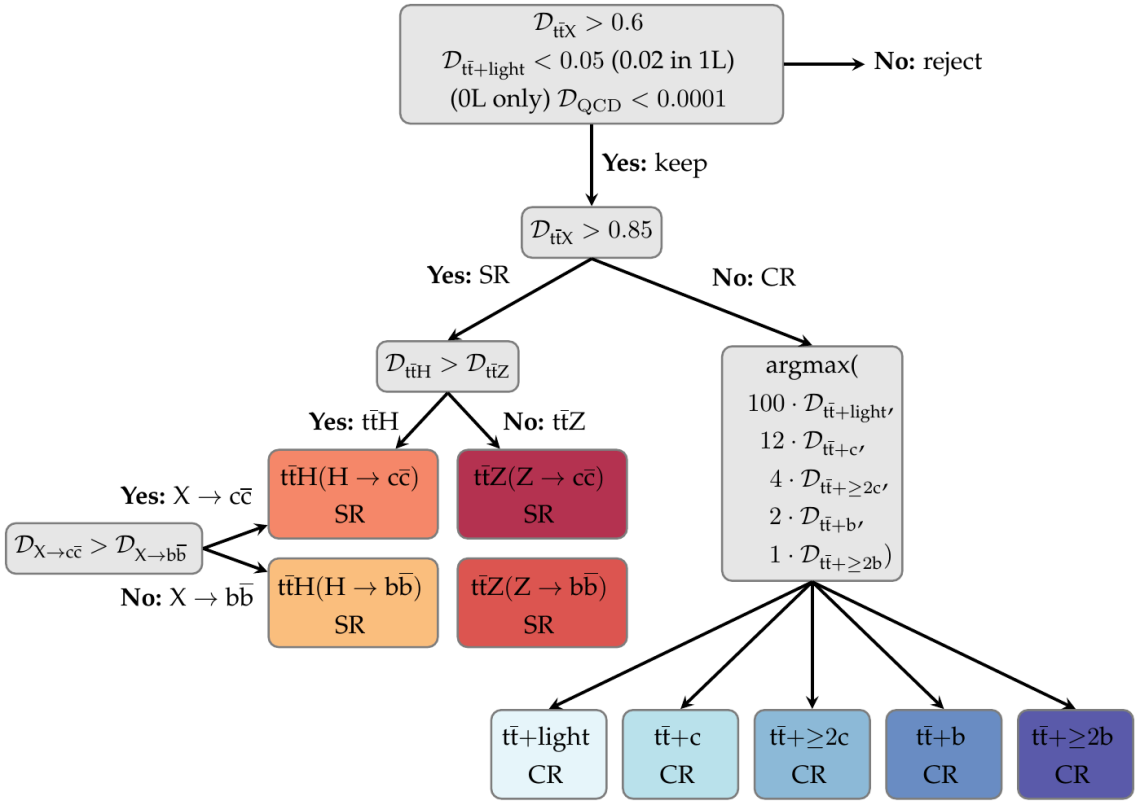
Signal, control and validation regions from one classifier

Event classifier: ParT (transformer) on jets, leptons and p_T^{miss}: one score per process (4 signals, 5 $t\bar{t}$+jets, + QCD in 0ℓ)

$D_{t\bar{t}X}$ = sum of the four signal scores

rejected	Validation	5 control regions	4 SRs
< 0.4	0.4–0.6	0.6–0.85	> 0.85

- **SRs**: $t\bar{t}H$ / $t\bar{t}Z$ × $c\bar{c}$ / $b\bar{b}$; c-enriched SRs need ≥ 2 c jets
- **CRs**: one per $t\bar{t}$+jets flavour; their normalisations float in the fit
- **Validation**: same structure, no signal: good agreement → backup
- **QCD** (0ℓ): a classifier cut leaves < 1% of $t\bar{t}$+jets
- **Fit**: ParT score in all regions of the 3 channels at once

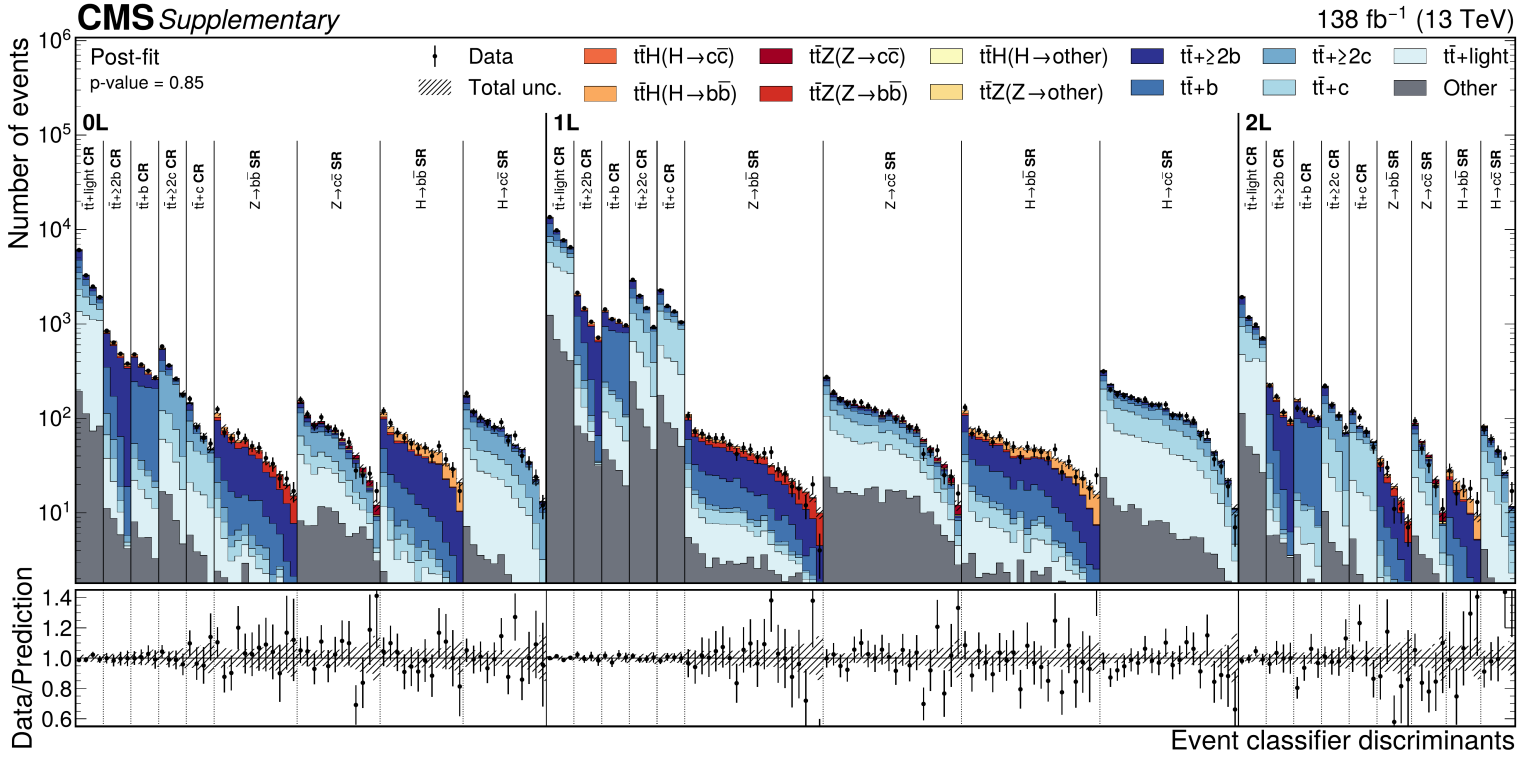


Categorisation of events with $D_{t\bar{t}X} > 0.6$ (same tree in the VR, 0.4–0.6)

$t\bar{t}H(H \to c\bar{c})$: results

PRL 136 (2026) 011801 · CMS-HIG-24-018

One fit to all regions: four signals and every $t\bar{t}$+jets component



Post-fit ParT discriminants in the 5 CRs and 4 SRs of each channel (0ℓ, 1ℓ, 2ℓ) · fit p-value 0.85

$t\bar{t}H(H \to c\bar{c})$ · 95% CL · observed (expected)

$\mu < 7.8$ (8.7)

$|\kappa_c| < 3.0$ (3.3)

$\kappa_b = 1$ · with κ_b fitted: $|\kappa_c| < 4.2$ obs. → backup

	best fit $\hat{\mu}$	obs. (exp.)
$t\bar{t}H(b\bar{b})$	$0.91\ ^{+0.26}_{-0.22}$	4.4σ (4.5σ)
$t\bar{t}H(c\bar{c})$	**−1.6 ± 4.5**	–
$t\bar{t}Z(b\bar{b})$	$1.47\ ^{+0.45}_{-0.41}$	3.5σ (2.4σ)
$t\bar{t}Z(c\bar{c})$	$1.02\ ^{+0.79}_{-0.84}$	1.2σ (1.3σ)

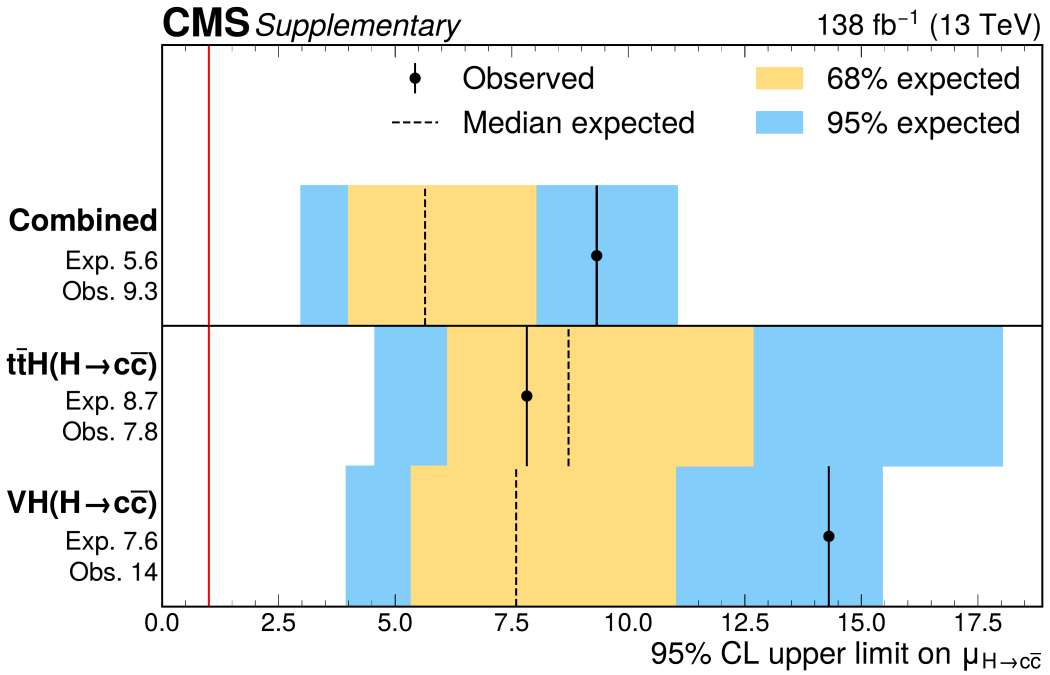
$t\bar{t}Z(c\bar{c})$ and $t\bar{t}Z(b\bar{b})$ are the standard candles: both consistent with the SM · limits per channel and event yields → backup

→ A second direct channel, with expected sensitivity comparable to VH

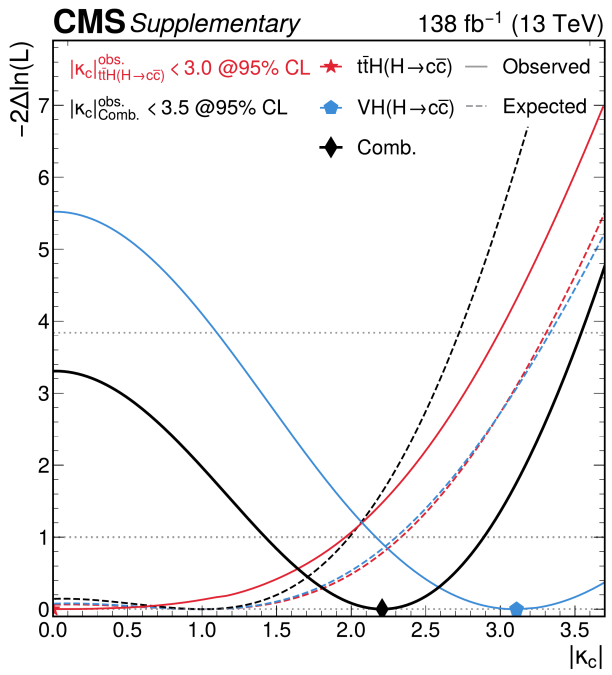
$t\bar{t}H$ + VH combination

PRL 136 (2026) 011801 · CMS-HIG-24-018

Best expected direct sensitivity to κ_c



95% CL limits on $\mu_{H\to c\bar{c}}$: $t\bar{t}H$, VH and combined



$-2\Delta\ln L$ vs $|\kappa_c|$ ($\kappa_b = 1$)

$t\bar{t}H$ + VH · 95% CL
observed (expected)

$\mu < 9.3$ (5.6)

$|\kappa_c| < 3.5$ (2.7)

SM production, $\kappa_b = 1$

26%
better expected μ limit than VH alone

36%
better expected μ limit than $t\bar{t}H$ alone

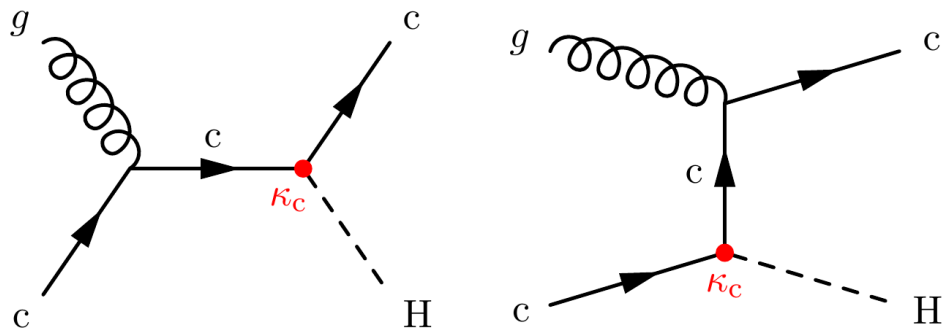
$|\kappa_c| < 2.7$
expected, below the BSM-motivated ~3

➜ The two direct channels together set the best expected constraint on κ_c

H + charm production

JHEP 11 (2025) 060 · JHEP 08 (2026) 051

κ_c in the production vertex, independent of the H → $c\bar{c}$ decay



LO diagrams for cH production; κ_c at the marked vertices

Signal: a clean Higgs decay (γγ or WW) + one c-tagged jet

Challenges

1 Tiny rate

σ_{cH} = 90 fb, about 0.2% of all Higgs production: $\sigma\mathcal{B}$ ≈ **0.2 fb** (γγ), ≈ 20 fb (WW)

2 Yukawa-insensitive processes

ggH and other Higgs processes with an extra c jet (e.g. from g → $c\bar{c}$) give the **same final state** without involving κ_c

3 b-associated Higgs (bH)

σ_{bH} = 660 fb, 7× larger: b jets mistagged as c, so c-vs-b separation matters

4 Modelling

Flavour scheme for cH (4FS vs 3FS): **25–30%** uncertainty; heavy-flavour modelling of ggH (T. Bevilacqua, next talk)

→ A complementary handle on κ_c, but small rates and irreducible H+c make it hard

JHEP 11 (2025) 060 · CMS-HIG-23-010

cH(H → γγ)

First search for H + charm in the diphoton channel, 138 fb^{-1}

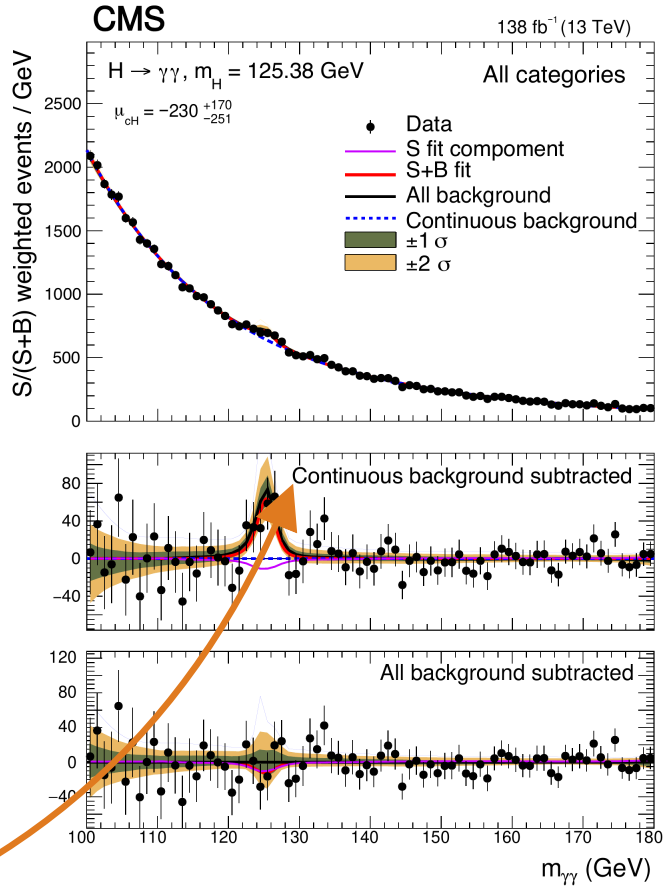
① Selection
- Two identified photons
- Leading jet (p_T > 25 GeV) **c-tagged** with DeepJet

② Two BDTs, no c-tag inputs
- cH vs **ggH**, the largest resonant background
- cH vs the **continuum** γγ + jets

③ Categories and fit
- 27 categories (9 per year)
- Signal from fits to $m_{\gamma\gamma}$

μ_{cH} < 243 (355) **$|\kappa_c|$ < 38** (73)

95% CL · observed (expected)



$m_{\gamma\gamma}$, all categories, S/(S+B) weighted

The 125 GeV peak is explained by the other Higgs processes (mostly ggH)

→ $|\kappa_c| \lesssim 40$: $\sigma\mathcal{B} \approx 0.2$ fb, and Yukawa-insensitive H+c looks the same

cH(H → WW) and the γγ + WW combination

JHEP 08 (2026) 051 · CMS-HIG-24-009

evμν + a c jet, then combined with γγ

① Selection

- One electron + one muon, p_T^{miss} > 20 GeV
- ≥ 1 **c jet** (DeepJet): 43% c efficiency, 27% b and 5% light mistag, to separate cH from bH

② Two BDTs

- vs non-Higgs backgrounds
- vs other Higgs processes; signal regions with 1 or > 1 c jets

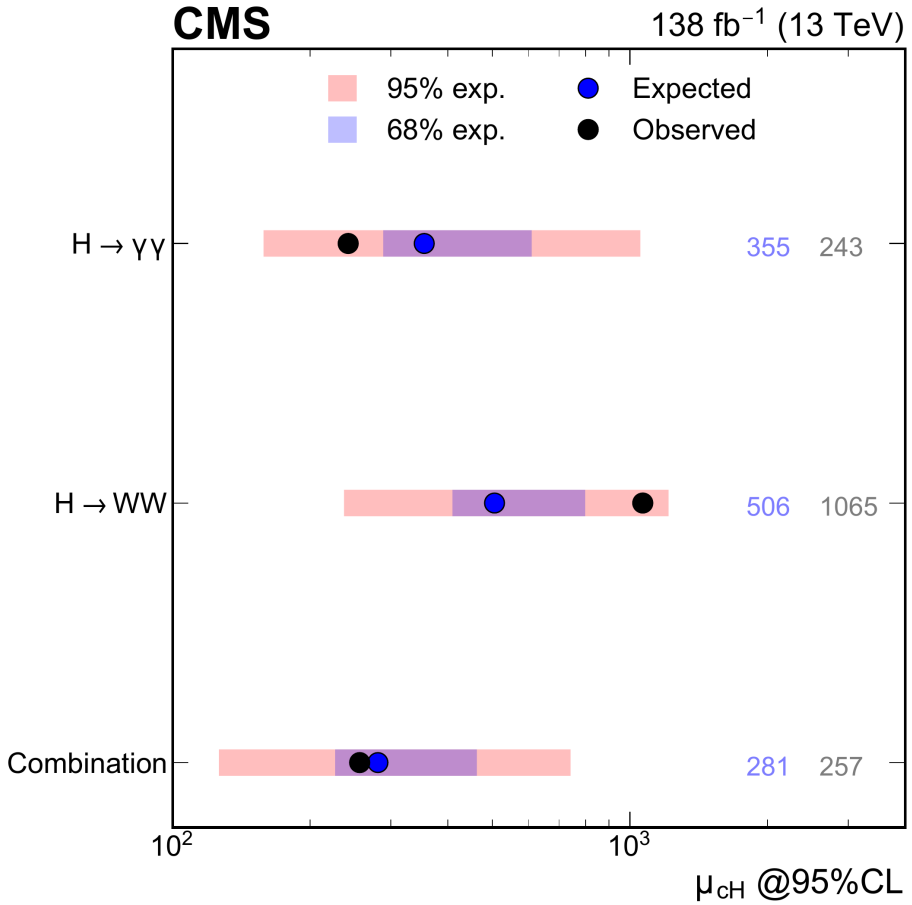
③ Fit

- μ_{cH} is the only parameter; all other Higgs processes fixed to their SM prediction (floating the H+c part → backup)

γγ + WW · 95% CL · observed (expected)

μ_{cH} < 257 (281) $|\kappa_c|$ < 47 (51)

WW alone: μ_{cH} < 1065 (506), $|\kappa_c|$ < 211 (95)



μ_{cH} limits: γγ, WW and combined

→ κ_c from production alone, independent of the H → $c\bar{c}$ decay; statistics-limited

Indirect constraints on κ_c

PRD 112 (2025) 112001 · arXiv:2607.23352

The Higgs width from on- and off-shell H → ZZ → 4ℓ

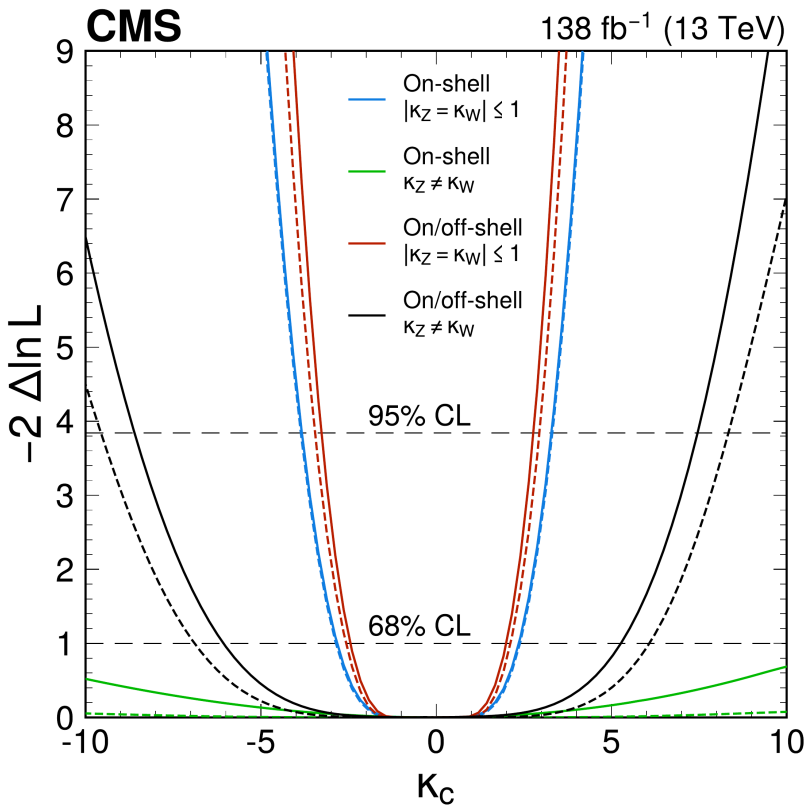
$$\sigma_{on} \propto \frac{g_p^2\, g_d^2}{\Gamma_H} \qquad \sigma_{off} \propto g_p^2\, g_d^2$$

g_p, g_d: production and decay couplings ($g_d \propto \kappa_Z$)

1. **Large κ_c** adds $c\bar{c}$ → H production and a larger width Γ_H
2. **On-shell** $\propto 1/\Gamma_H$: larger κ_Z can hide larger Γ_H ⇒ need $|\kappa_Z| \leq 1$
3. **Off-shell** has no Γ_H ⇒ fixes κ_Z, assumption dropped

κ_c at 95% CL	observed	expected
On-shell, $\lvert\kappa_Z = \kappa_W\rvert \leq 1$	[−3.7, 3.2]	[−3.8, 3.3]
new On + off-shell, $\lvert\kappa_V\rvert \leq 1$	**[−3.3, 2.8]**	[−3.5, 2.9]
new On + off-shell, κ_Z, κ_W free	[−8.6, 7.5]	[−9.6, 8.3]

Higgs width: F. Errico (Tue) · p_T^H spectrum → backup



On-shell (blue, green), on + off-shell (red, black); solid observed, dashed expected

→ With $|\kappa_V| \leq 1$, H → 4ℓ matches the direct searches, under different assumptions

Where we stand

CMS constraints on κ_c at 95% CL

Probe	Reference	observed	expected
$t\bar{t}H$ + $VH(H \to c\bar{c})$	PRL 136 (2026) 011801	**$\lvert\kappa_c\rvert < 3.5$**	2.7
$t\bar{t}H(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.0$	3.3
$VH(H \to c\bar{c})$	PRL 131 (2023) 061801	$1.1 < \lvert\kappa_c\rvert < 5.5$	3.4
On + off-shell $H \to 4\ell$, $\lvert\kappa_V\rvert \leq 1$	arXiv:2607.23352	**[−3.3, 2.8]**	[−3.5, 2.9]
On-shell $H \to 4\ell$, $\lvert\kappa_Z = \kappa_W\rvert \leq 1$	PRD 112 (2025) 112001	[−3.7, 3.2]	[−3.8, 3.3]
On + off-shell $H \to 4\ell$, κ_Z, κ_W free	arXiv:2607.23352	[−8.6, 7.5]	[−9.6, 8.3]
cH ($\gamma\gamma$ + WW)	JHEP 08 (2026) 051	$\lvert\kappa_c\rvert < 47$	51
$H \to J/\psi\,\gamma$ · sign-sensitive · J. Bierkens, Thu	PLB 865 (2025) 139462	[−166, 208]	[−128, 169]

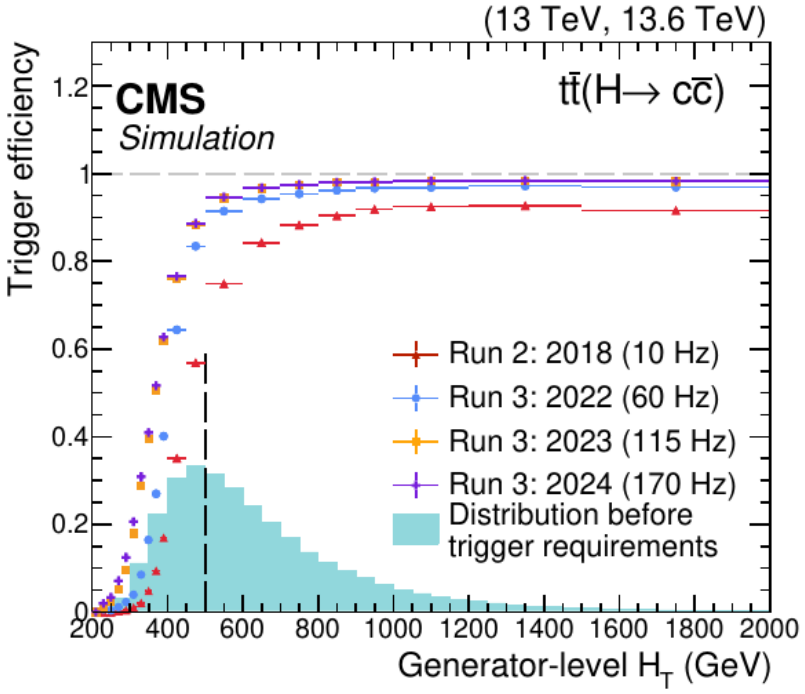
The results use different assumptions on the other Higgs couplings and are not directly comparable

→ Direct searches and on + off-shell $H \to 4\ell$ both reach $\lvert\kappa_c\rvert \sim 3$, where flavour-compatible BSM lives

Toward Run 3

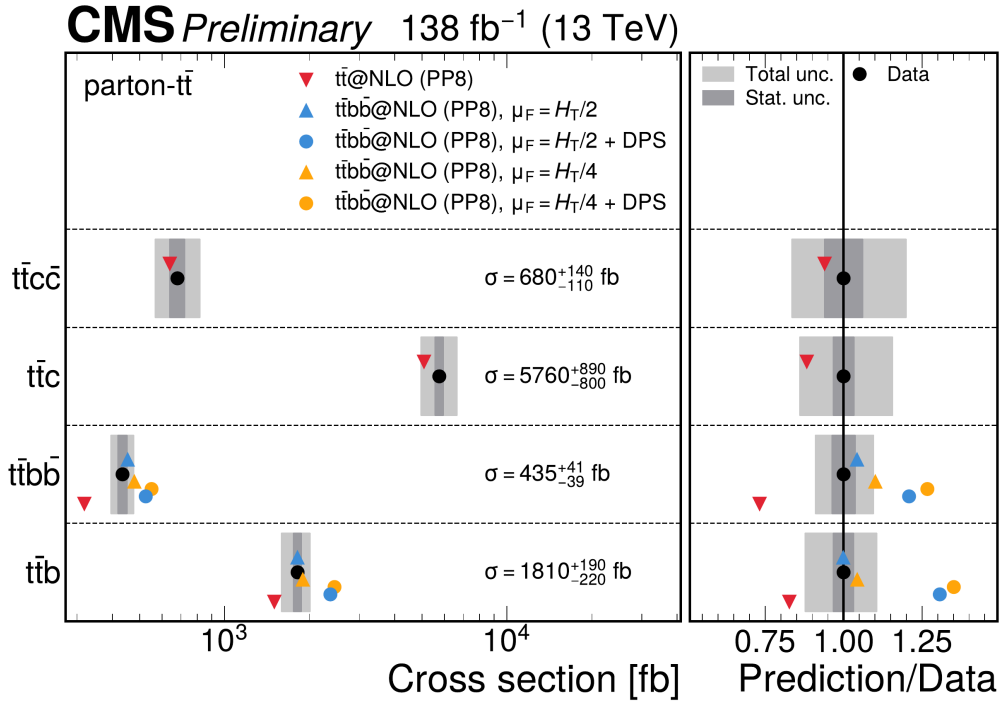
Better triggers, and the main $t\bar{t}H$ background measured

Run 3: ML b/c tagging at the HLT



arXiv:2608.18331 (BTV-25-002), simulation: $t\bar{t}H(H \to c\bar{c})$ trigger efficiency, Run 2 (2018) vs Run 3 (2022–24), 13.6 TeV · HLT taggers: J. Daniel (Mon) → backup

Run 2: $t\bar{t}c\bar{c}$ cross section measured



CMS-PAS-TOP-26-003 **new**, 138 fb^{-1}: $\sigma_{t\bar{t}c\bar{c}}$ = 680 ± 40 (stat) $^{+130}_{-100}$ (syst) fb

➜ Run 3 starts with higher trigger efficiency and a $t\bar{t}$+$c\bar{c}$ background measured in Run 2

Run 3 and HL-LHC prospects

arXiv:2504.00672 · CMS-HIG-25-002

From $|\kappa_c| \sim 3$ toward κ_c of order one

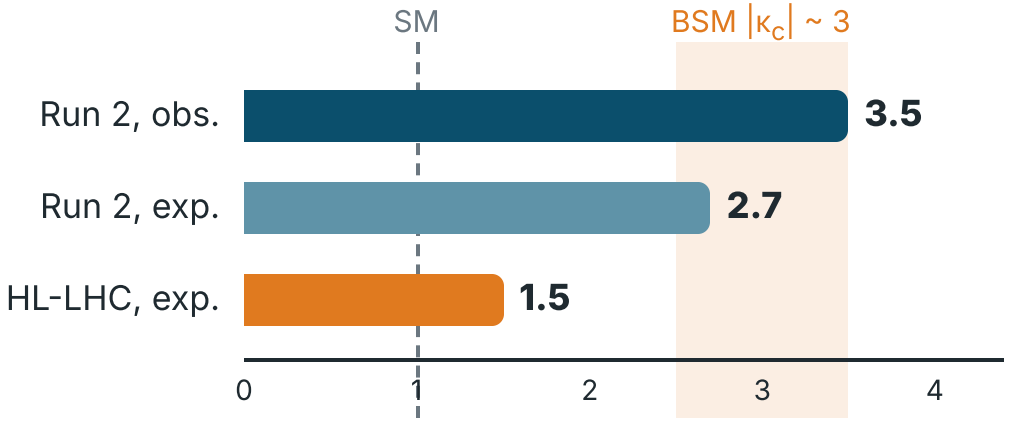
Run 3 (13.6 TeV)

- **Statistics-limited**: more data helps directly
- Better c tagging: UParT (slide 3)
- Better triggers (slide 17)

HL-LHC: 3000 fb^{-1} per experiment (≈ 20× Run 2)

- **Expected $|\kappa_c| < 1.5$**, ATLAS + CMS combined
- VH(H → $c\bar{c}$): ≈ 1.6σ expected, still no observation
- 30% better than the previous projection, thanks to better b and c tagging

Direct searches: allowed $|\kappa_c|$



Run 2: PRL 136 (2026) 011801 · HL-LHC: arXiv:2504.00672
No bar excludes $\kappa_c = 0$ yet

κ_c of order one: first sensitivity to whether the Higgs gives charm its mass

Summary

Does the Higgs give charm its mass?

The charm Yukawa is the most accessible second-generation quark coupling; CMS probes it **in the decay, in production and indirectly**

Direct

$t\bar{t}H$ + VH($H \to c\bar{c}$)

$|\kappa_c| < 3.5$ (2.7)

most stringent direct constraint, best expected sensitivity

Production

H + charm, γγ + WW

$|\kappa_c| < 47$ (51)

independent of the $H \to c\bar{c}$ decay

Indirect

on + off-shell $H \to 4\ell$

$[-3.3, 2.8]$

with $|\kappa_V| \leq 1$ (about ±8 without)

Next

Run 3 data

better taggers and triggers

$t\bar{t}c\bar{c}$ background already measured

Today **$|\kappa_c| \lesssim 3$–3.5**: entering BSM-motivated territory

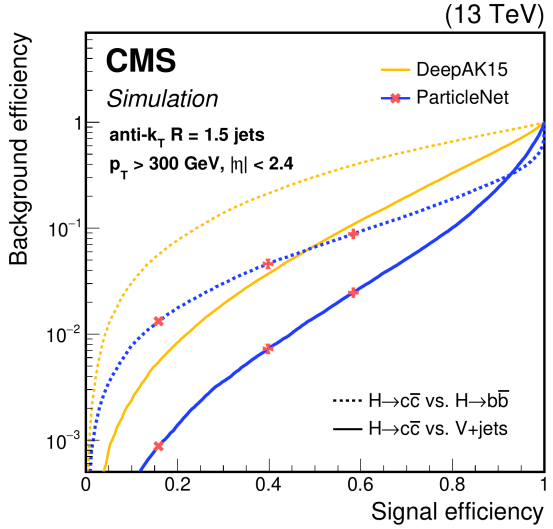
HL-LHC expected **$|\kappa_c| < 1.5$**: approaching, not yet reaching, sensitivity to $\kappa_c = 0$

➜ Thank you!

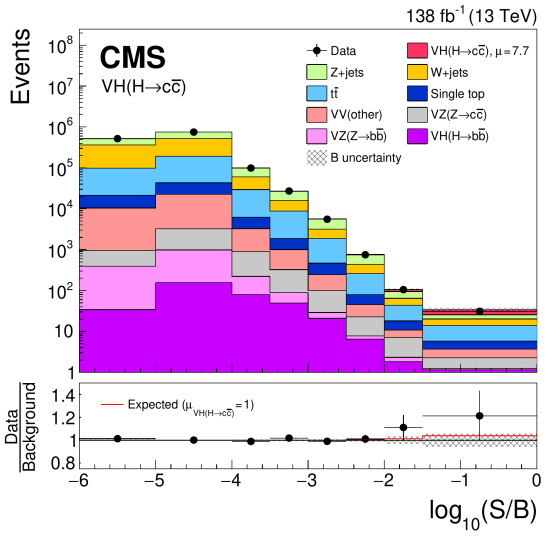
Backup

VH(H → $c\bar{c}$): categories and sensitivity

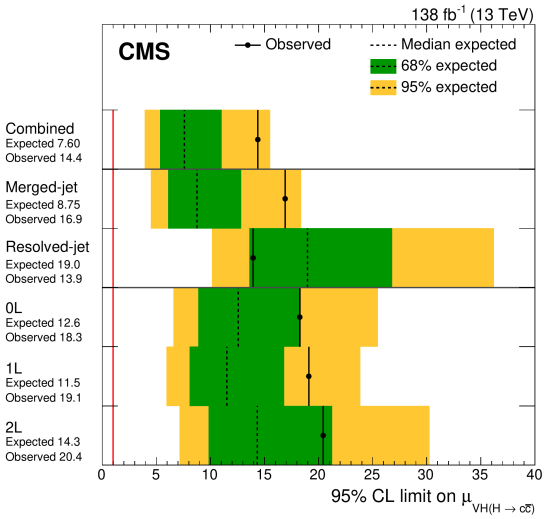
PRL 131 (2023) 061801



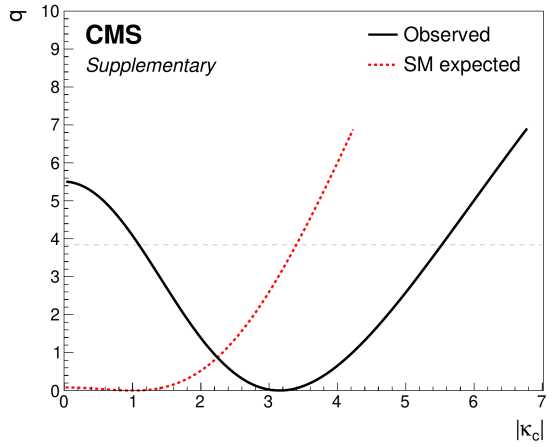
ParticleNet merged-jet H → $c\bar{c}$ tagging



Events vs $\log_{10}$(S/B)



μ limits per channel (inner band 68%)



|κ_c| scan (supplementary)

Best fit μ = 7.7 $^{+3.8}_{-3.5}$; merged μ = 8.7 $^{+4.6}_{-4.0}$, resolved μ = −9.5 ± 9.6 · σ$\mathscr{B}$ < 0.94 (0.50) pb · μ limits with CLs, κ_c intervals from profile-likelihood scans

➜ *The upward fluctuation comes from the merged-jet category*

VH(H → $c\bar{c}$): analysis details

PRL 131 (2023) 061801 · strategy as in JHEP 03 (2020) 131

Topologies Two, orthogonal: H candidate from a large-R (R = 1.5) jet with p_T above (merged) or below (resolved) 300 GeV; same lepton channels

Resolved DeepJet CvsL/CvsB; the two highest-CvsL jets form the H candidate; leading-jet cuts at ≈40% c efficiency (≈4% light, ≈16% b mistag). c-jet energy regression, FSR-jet recovery and a 2ℓ kinematic fit improve the $m(H_{cand})$ resolution by up to 20%. BDT per channel, fit to the BDT score

Merged ParticleNet $c\bar{c}$ tagger, 4–7× better rejection than DeepAK15; 3 working points (≈58, 40, 16% $c\bar{c}$ efficiency) define 3 purity categories. ParticleNet mass regression: ≈50% better mass resolution. BDTs decorrelated from $m(H_{cand})$; fit to $m(H_{cand})$

Backgrounds V+jets and $t\bar{t}$ normalised in dedicated control regions, following the 2016 analysis (V+jets split by jet flavour, free rate parameters)

VZ validation Same analysis, resolved BDTs retrained with VZ as signal, merged unchanged. $\mu_{VZ} = 1.01\ ^{+0.23}_{-0.21}$; 4.4σ merged, 3.1σ resolved, 5.7σ combined (exp. 4.7, 3.3, 5.9)

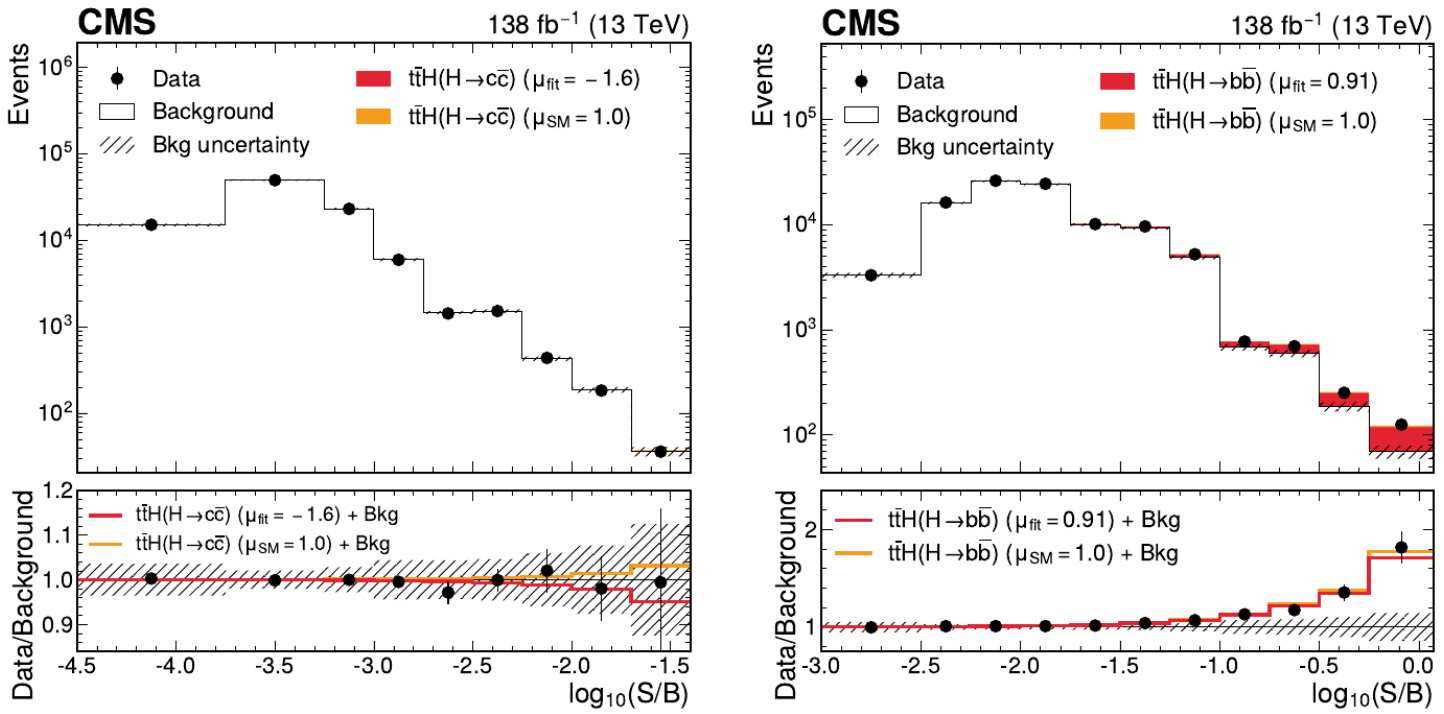
Source	$\Delta\mu/\Delta\mu_{tot}$
Statistical	**85%**
Bkg. normalisations	37%
Experimental	48%
Simulated sample sizes	37%
c-jet identification	23%
Jet energy scale, resolution	15%
Simulation modelling	11%
Luminosity	6%
Lepton identification	4%
Theory	22%
Backgrounds	17%
Signal	15%

For $\hat{\mu} = 7.7\ ^{+3.8}_{-3.5}$; simulation size driven by negative weights in NLO V+jets

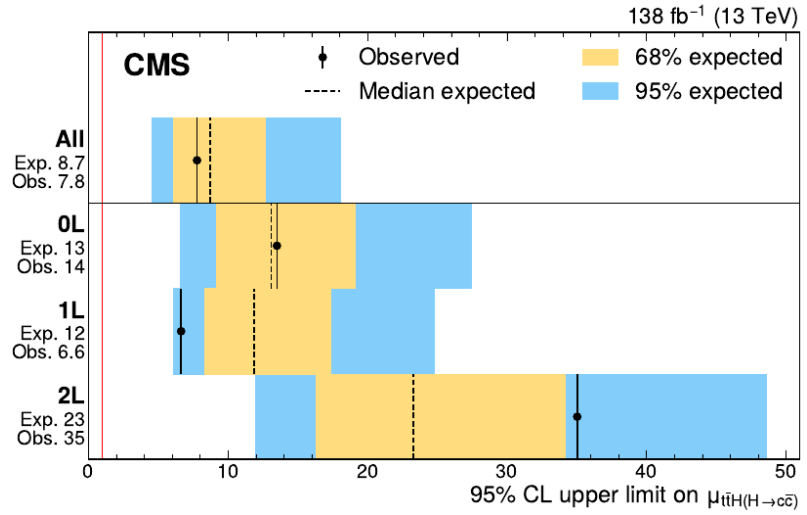
➜ Statistics-limited; c tagging and simulation size come next

$t\bar{t}H(H \to c\bar{c})$: event yields

PRL 136 (2026) 011801



Event yields vs $\log_{10}(S/B)$; the $t\bar{t}H(c\bar{c})$ panel shows the published best fit



$\mu_{ttH(c\bar{c})}$ limits per channel (inner band 68%)

→ *$t\bar{t}H(c\bar{c})$ best fit $\hat{\mu} = -1.6 \pm 4.5$: compatible with zero and with the SM*

$t\bar{t}H(H \to c\bar{c})$: flavour tagging and by-products

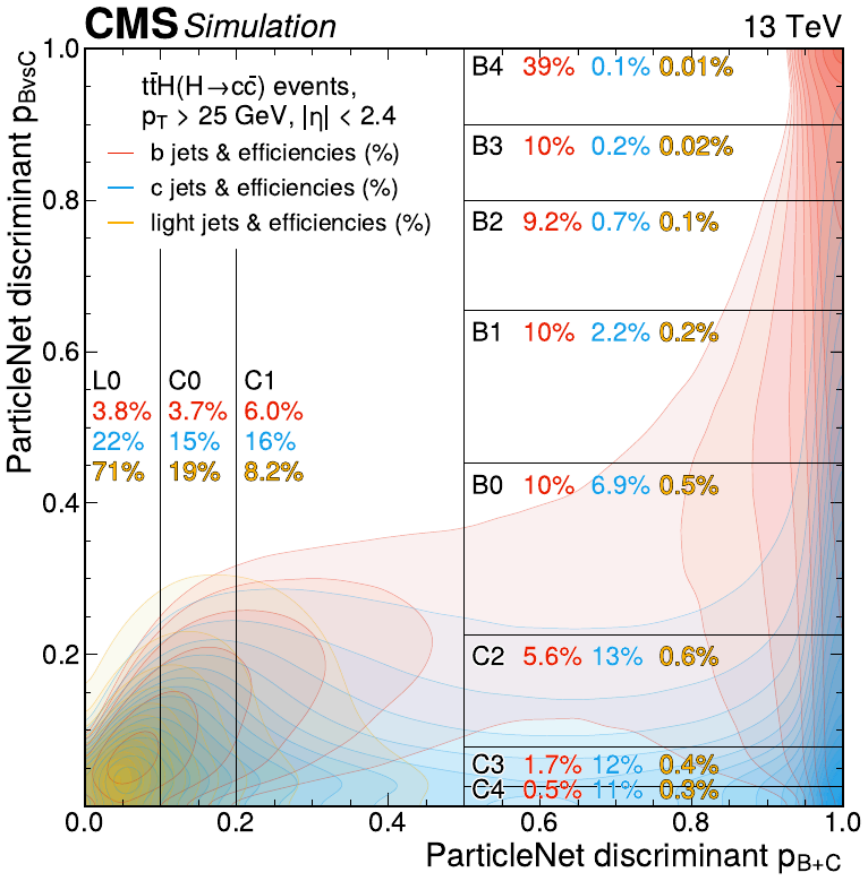
PRL 136 (2026) 011801

Tagging

- Each jet gets ParticleNet scores; the (b+c vs light, b vs c) plane is split into 2D categories
- ParT event classifier separates $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$, $t\bar{t}$ +HF

By-products

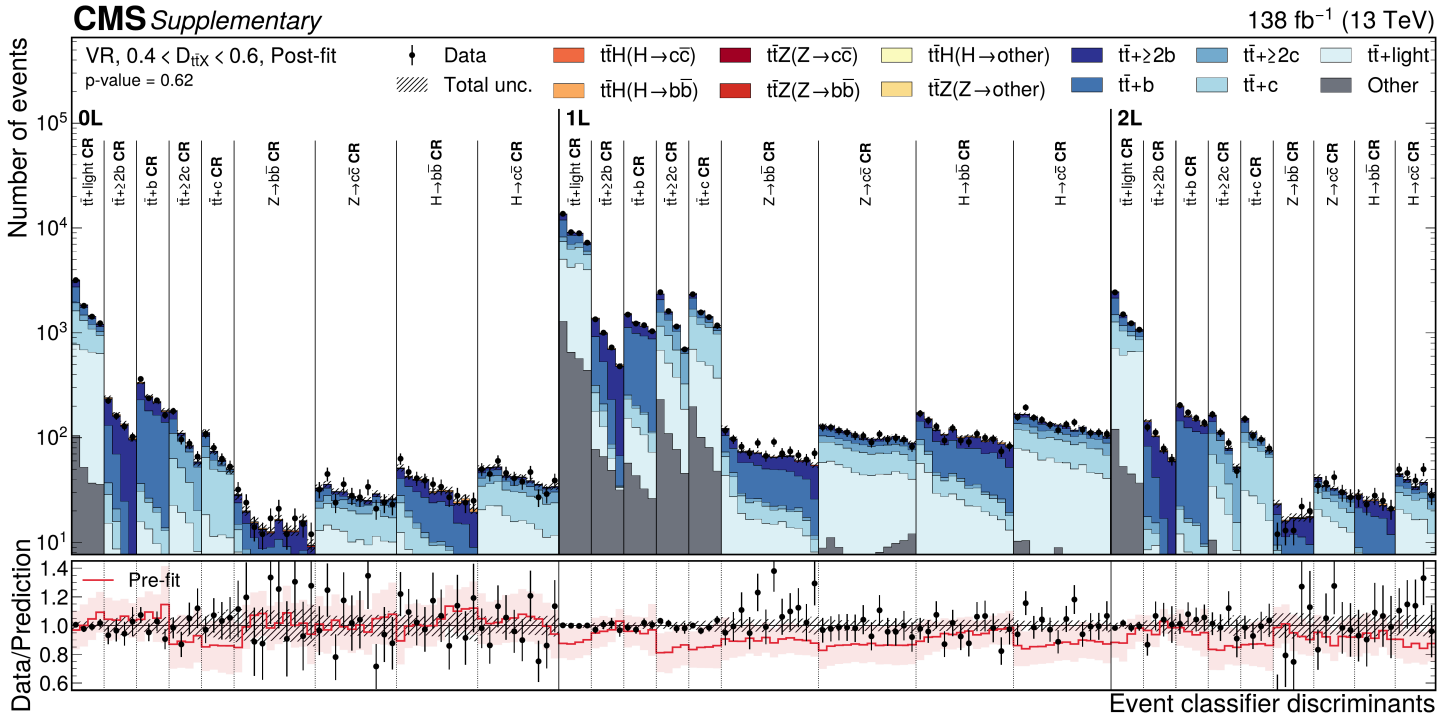
- $t\bar{t}Z(Z \to c\bar{c})$ standard candle: $\mu = 1.0 \pm 0.8$
- $t\bar{t}H(H \to b\bar{b})$: $\mu = 0.91\ ^{+0.26}_{-0.22}$, 4.4σ (4.5σ exp.)



b, c, light jets in the ParticleNet plane and the 2D tagging categories

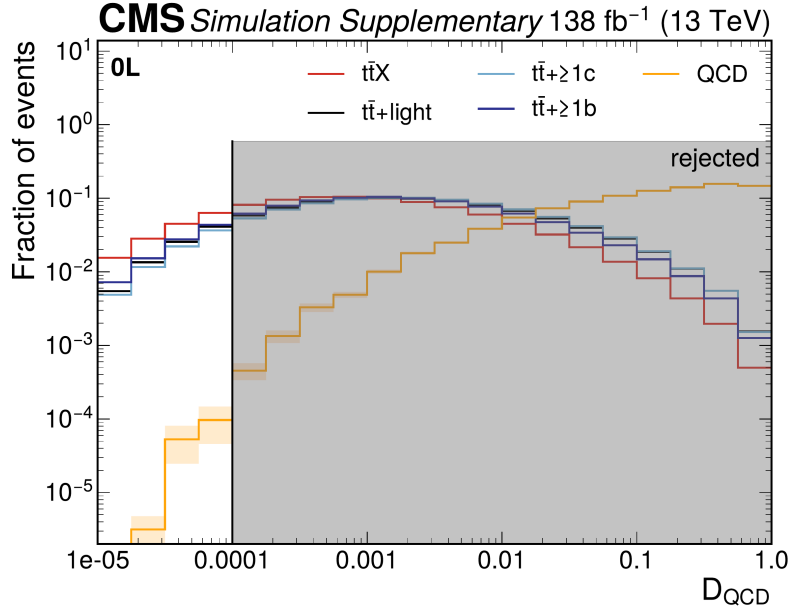
$t\bar{t}H$: validation region and QCD rejection

PRL 136 (2026) 011801 (supplementary)



Validation region 0.4 < $D_{t\bar{t}X}$ < 0.6, same SR/CR structure (CR-like 0.40–0.58, SR-like 0.58–0.60): good agreement

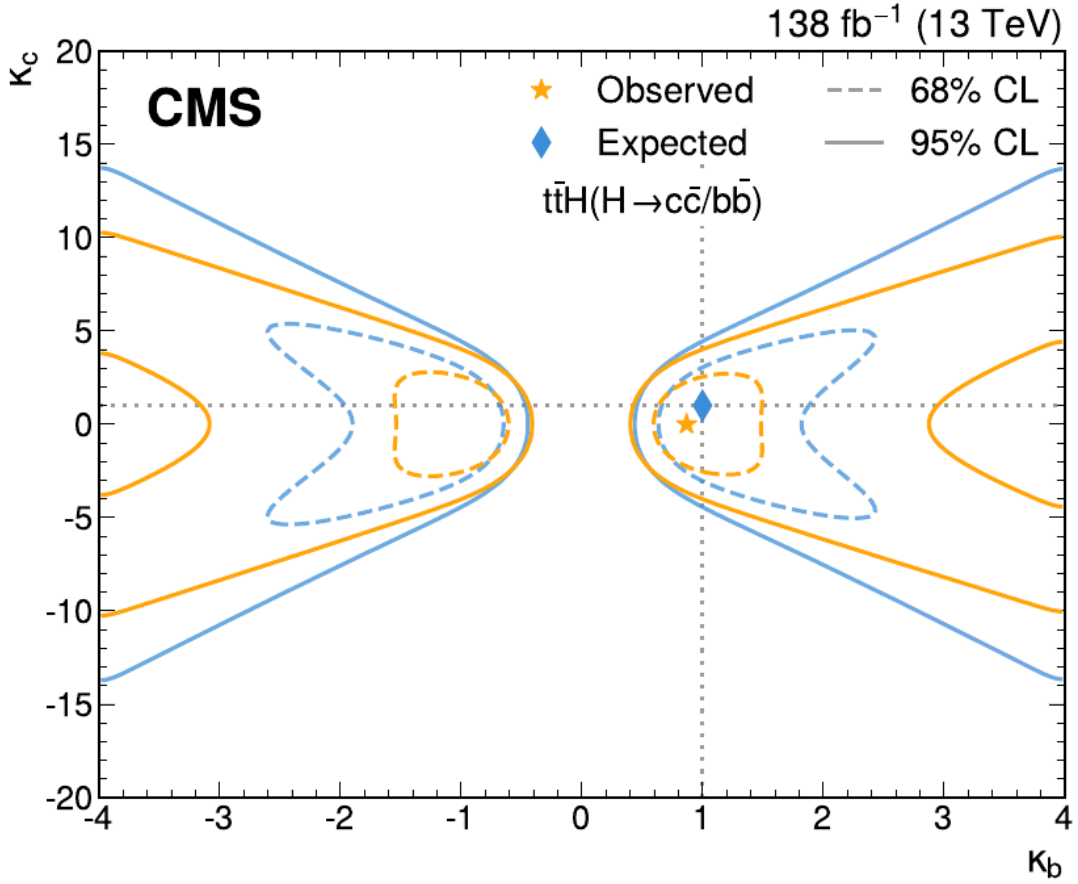
- c-enriched regions need ≥2 c jets ($t\bar{t}$+c: ≥1); CR choice weights 100, 12, 4, 2, 1 on the $t\bar{t}$+light, +c, +≥2c, +b, +≥2b scores
- $t\bar{t}$+b, +≥2b from a dedicated 4FS $t\bar{t}b\bar{b}$ sample



0ℓ only: D_{QCD} < 10^{-4} (grey area rejected) suppresses QCD by nearly four orders of magnitude, to <1% of $t\bar{t}$+jets; QCD simulation only trains the classifier

$t\bar{t}H$: the (κ_b, κ_c) plane

PRL 136 (2026) 011801



orange: observed, blue: expected; dashed 68%, solid 95% CL

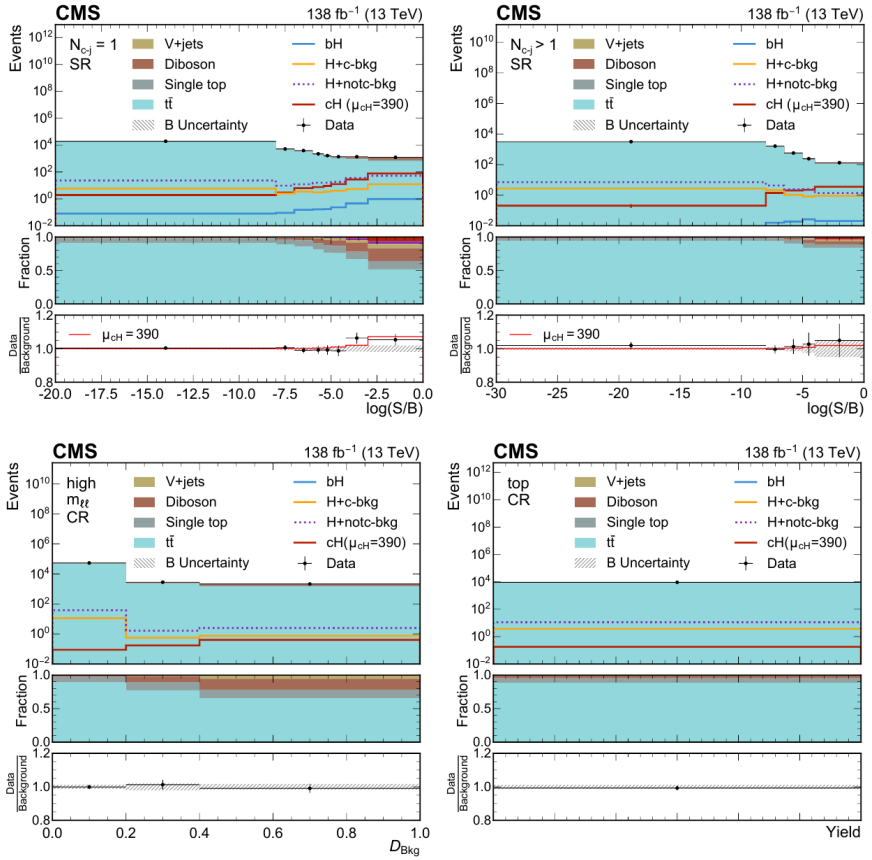
The simultaneous $t\bar{t}H(H \to b\bar{b})$ + $t\bar{t}H(H \to c\bar{c})$ fit constrains both Yukawas

Signal strengths: $t\bar{t}H(b\bar{b})$ $0.91\ ^{+0.26}_{-0.22}$ · $t\bar{t}H(c\bar{c})$ -1.6 ± 4.5 · $t\bar{t}Z(b\bar{b})$ $1.47\ ^{+0.45}_{-0.41}$ · $t\bar{t}Z(c\bar{c})$ $1.02\ ^{+0.79}_{-0.84}$

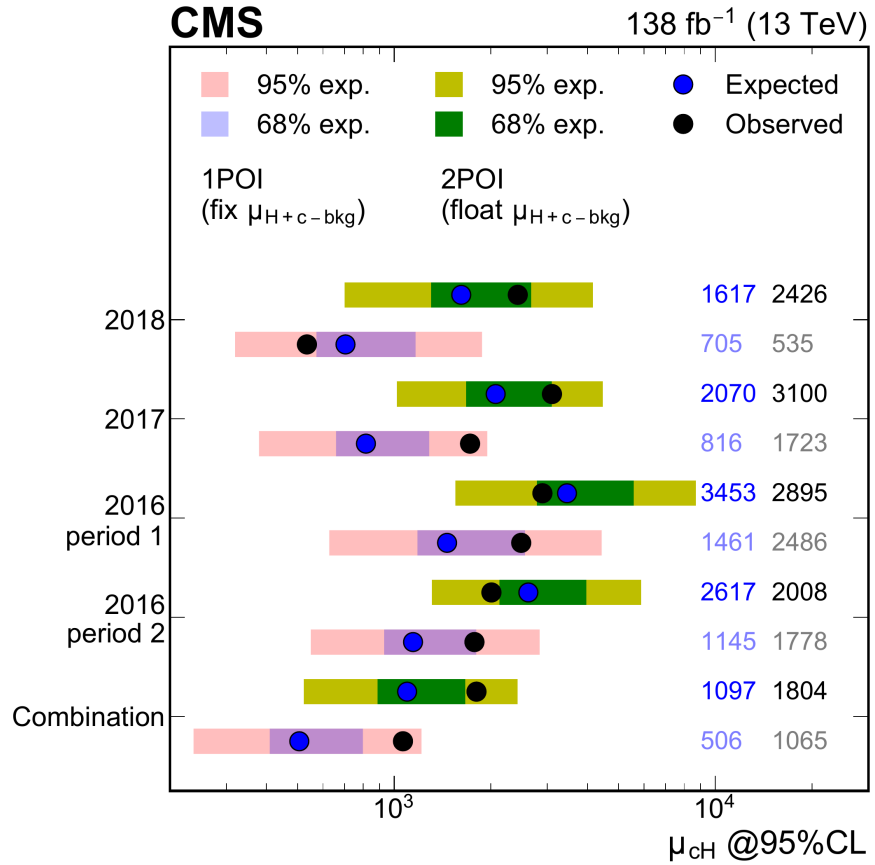
$|\kappa_c| < 4.2$ obs. with κ_b fitted

cH(H → WW): post-fit and per-period limits

JHEP 08 (2026) 051



1-POI post-fit BDT discriminants



μ_{cH} limits per data-taking period, 1 and 2 POI

➜ *Best fit (1 POI): $\mu_{cH} = 390\ ^{+380}_{-270}$, within 1.5σ of the SM; statistics-limited, then flavour-scheme and ggH heavy-flavour modelling*

cH(H → WW): floating the H+c background

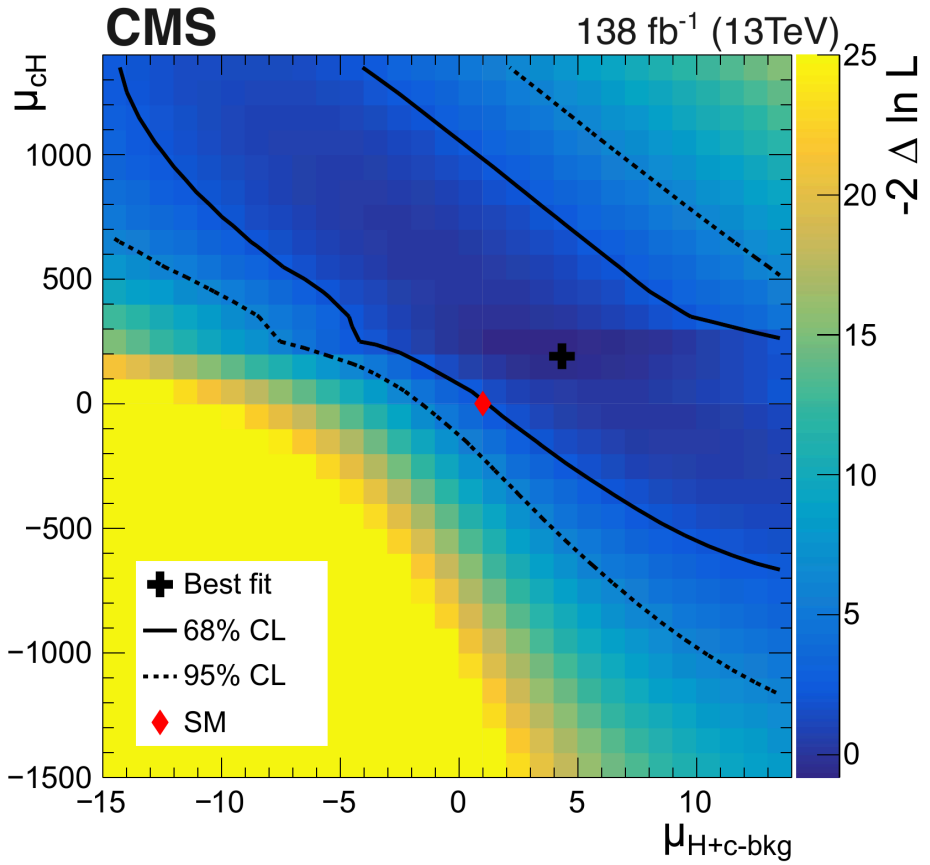
JHEP 08 (2026) 051

1 POI only μ_{cH} floats; all other Higgs processes (ggH, VBF, VH, $t\bar{t}H$, bH) are fixed to their SM prediction, with a 50% uncertainty on their heavy-flavour part

2 POI the charm-associated part, $\mu_{H+c\ bkg}$, also floats; correlation with cH: −87%

2-POI limit: $\mu_{cH} < 1804$ (1097)

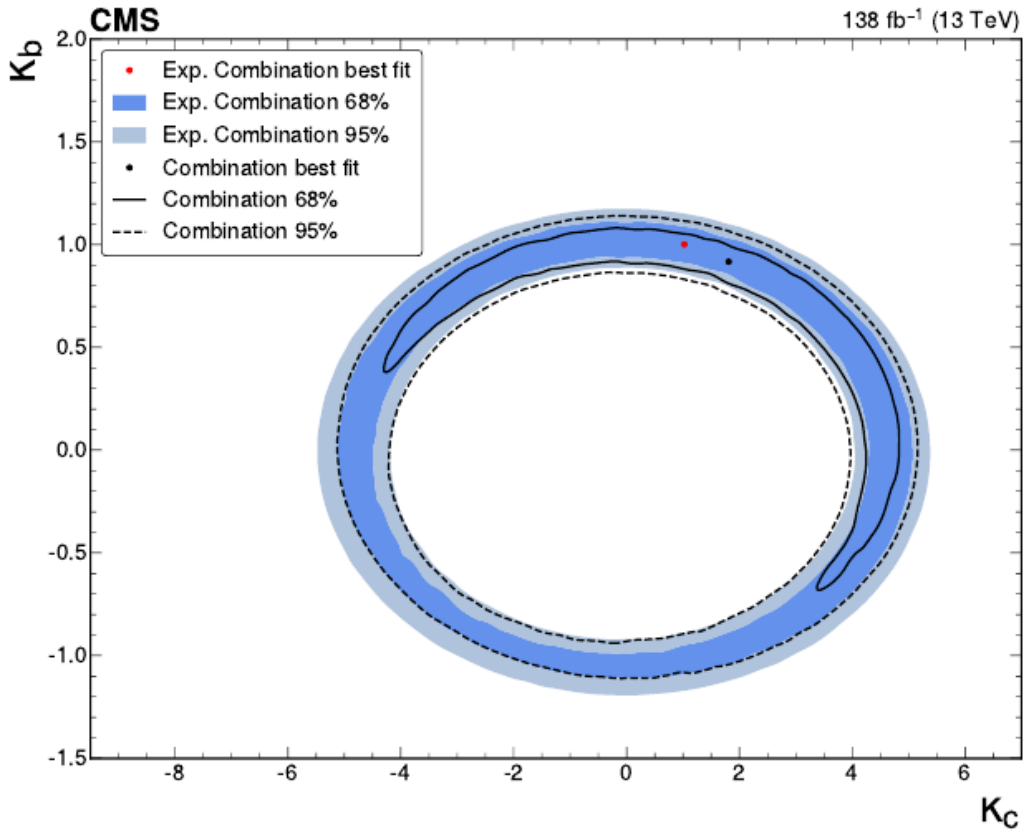
The combination with γγ uses the 1-POI fit



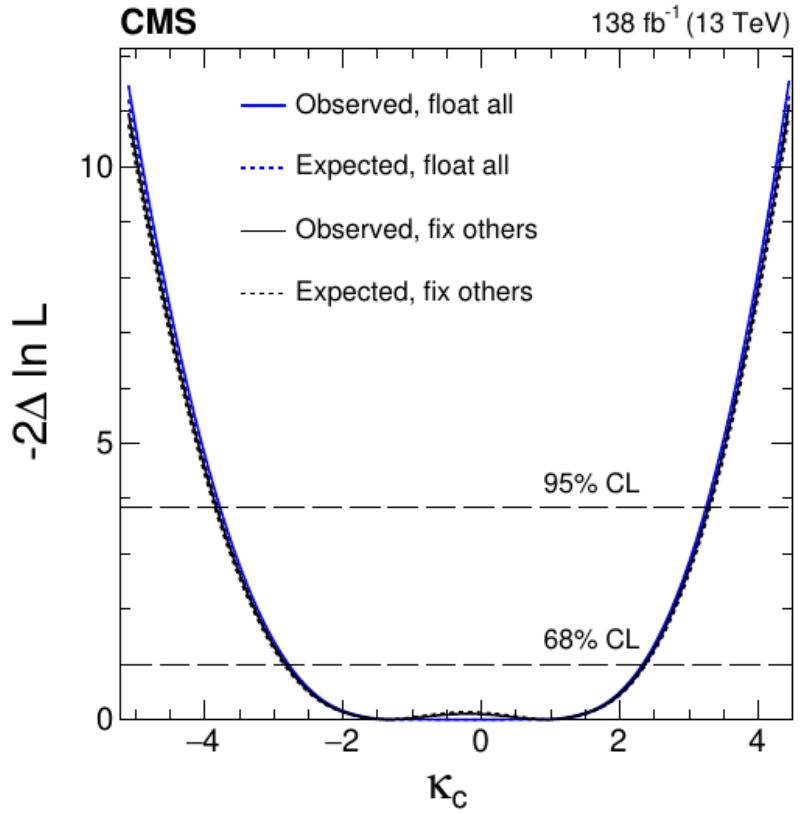
WW: μ_{cH} vs $\mu_{H+c\ bkg}$

➜ *Floating the H+c background weakens the WW limit from 1065 to 1804*

Higgs p_T spectrum and on-shell H → 4ℓ



JHEP 04 (2026) 093: (κ_b, κ_c) from the Run 2 p_T^H combination (γγ, ZZ, resolved ττ), $\mathscr{B}$ coupling-dependent; 2D contours only



PRD 112 (2025) 112001 (γH + on-shell H → 4ℓ): κ_c scan; solid obs., dashed exp.

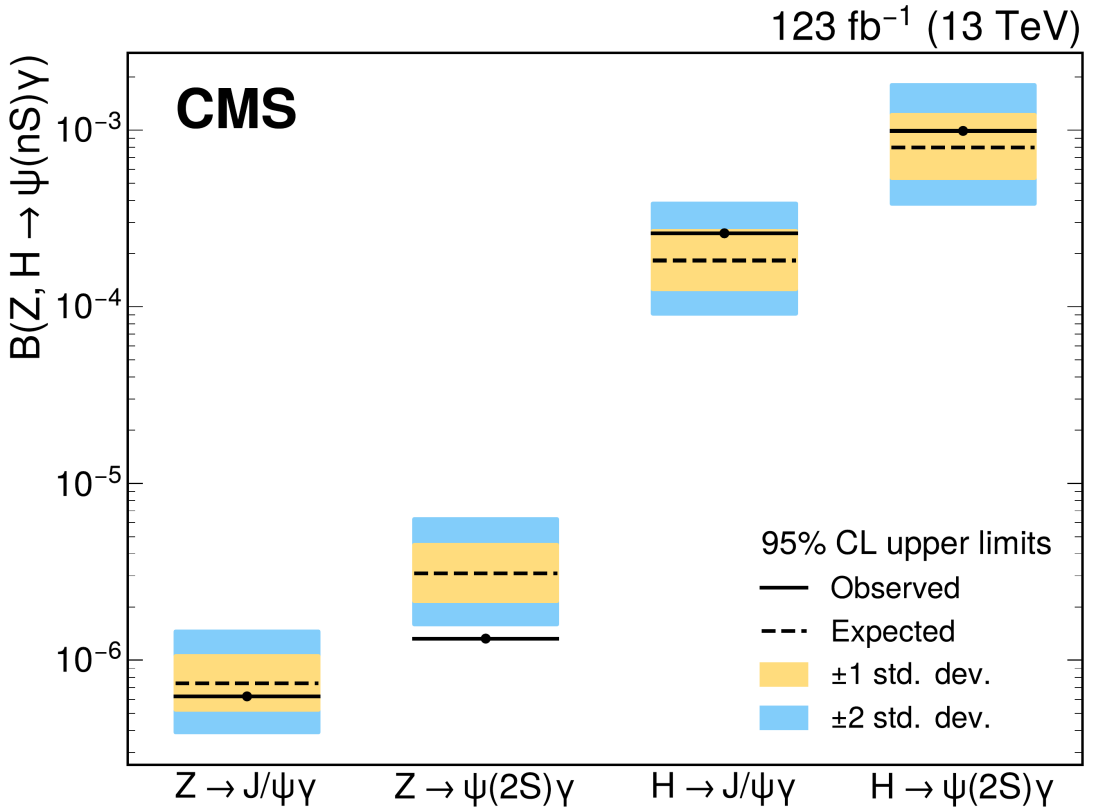
Rare exclusive decays: H → J/ψ γ, ψ(2S) γ

PLB 865 (2025) 139462 · CMS-SMP-22-012

Direct ($\propto \kappa_c$) and indirect ($H \to \gamma\gamma^* \to \gamma\, J/\psi$) amplitudes interfere ⇒ **sign sensitivity**

123 fb^{-1}: $\mathcal{B}(H \to J/\psi\,\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(H \to \psi(2S)\,\gamma) < 9.9 \times 10^{-4}$

$\kappa_c/\kappa_\gamma \in (-157, 199)$
with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$
expected (−128, 169) · 95% CL



95% CL upper limits on $\mathcal{B}(Z, H \to \psi(nS)\,\gamma)$, observed and expected

H → μμ: a second-generation Yukawa

JHEP 01 (2021) 148 · CMS-HIG-19-006

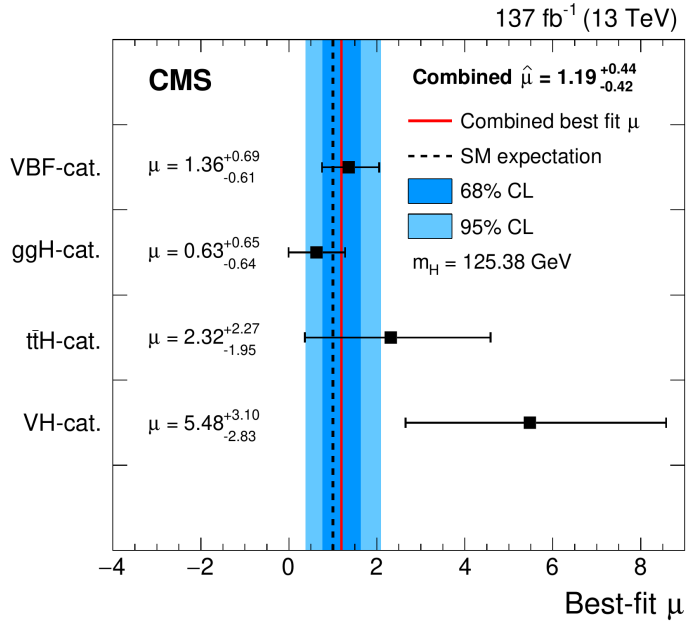
137 fb^{-1}, categories ggH, VBF, VH, $t\bar{t}H$

Excess of **3.0σ** (2.5σ exp.): first evidence for H decays to second-generation fermions

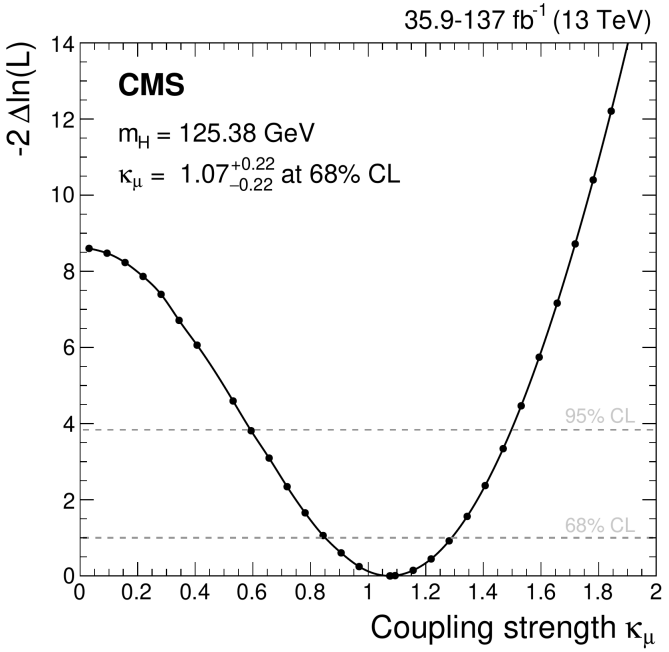
$\mu = 1.19\ ^{+0.40}_{-0.39}\ (\text{stat})\ ^{+0.15}_{-0.14}\ (\text{syst})$

$\kappa_\mu = 1.07$; 95% CL: $0.59 < \kappa_\mu < 1.50$ (6 free κ, SM total width)

Rare decays incl. μμ: L. D'Eramo's plenary (Wed)



μ̂ per production category and combined

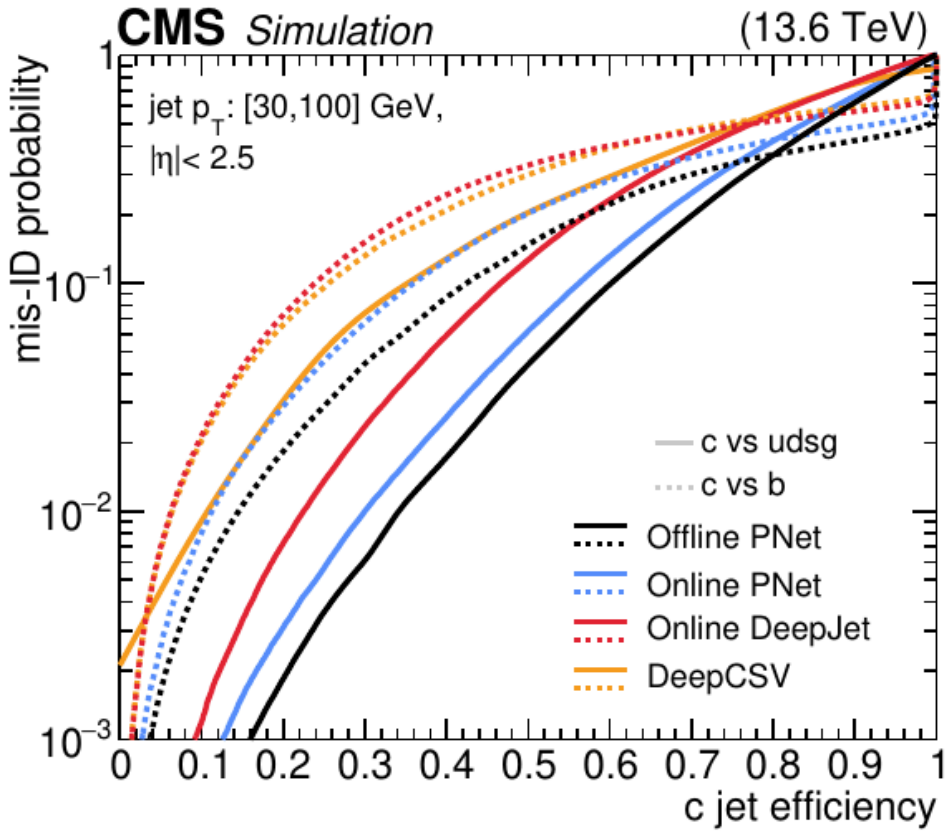


$-2\Delta\ln L$ vs κ_μ

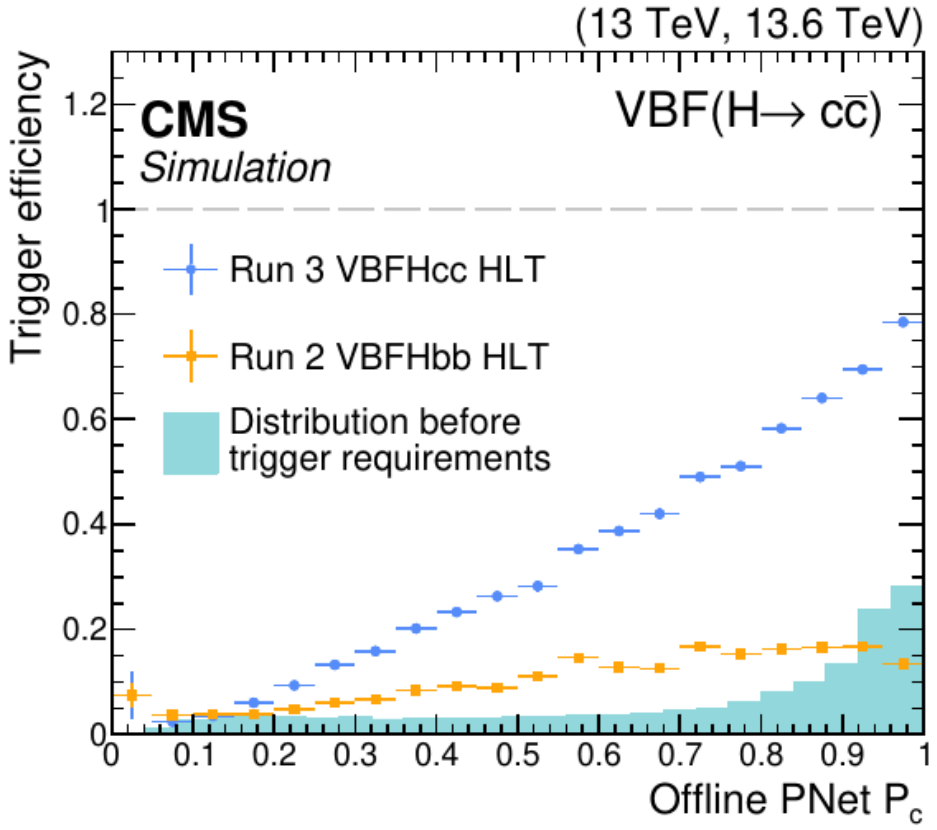
➜ The muon Yukawa is SM-like within about 20% (68% CL); charm is the next test

Run 3 HLT flavour tagging

arXiv:2608.18331 · CMS-BTV-25-002



Simulation, jet p_T 30–100 GeV: light (solid) and b (dashed) misidentification vs c-jet efficiency, online vs offline taggers



Simulation: VBF H → cc̄ trigger efficiency vs offline ParticleNet c score, Run 3 VBF H → cc̄ vs Run 2 VBF H → bb̄ trigger

Likely questions

- **How large is the VH excess?** Best fit $\mu_{VH} = 7.7\ ^{+3.8}_{-3.5}$, about 1.9σ above μ = 1 (estimated from the quoted uncertainty; the paper calls it a small upward fluctuation). This is why VH alone excludes $|\kappa_c| < 1.1$ and why $t\bar{t}H$ alone (3.0) beats the combination (3.5) on observed
- **What limits the direct searches?** Statistics first. $t\bar{t}H(c\bar{c})$: statistical 74% of the total uncertainty, then $t\bar{t}$+c and $t\bar{t}$+light modelling (≈32%, ≈29%). $VH(c\bar{c})$: simulation sample size (≈37%), c-tagging efficiency (≈23%), theory (≈22%)
- **And ATLAS?** VBF + VH(H → $c\bar{c}$) (arXiv:2511.21911): $|\kappa_c| < 4.7$ (3.9 exp.), same assumptions as CMS (other couplings SM, no BSM decays); its VBF part includes 51.5 fb^{-1} of Run 3 data
- **Why is cH an order of magnitude weaker than H → $c\bar{c}$?** cH $\sigma\mathscr{B}$ is tiny (≈0.2 fb in γγ) and ggH produces the same H+c final state independently of κ_c; in H → $c\bar{c}$, κ_c changes the branching fraction of Higgs bosons from much larger production modes (VH, $t\bar{t}H$)
- **Why is cH(γγ) obs. (38) much better than exp. (73)?** The best fit is negative, $\hat{\mu}_{cH} = -230\ ^{+170}_{-251}$
- **Why no κ_c from boosted H → $c\bar{c}$?** The paper quotes only μ; the result is ggH-dominated
- **How much of the Run 3 trigger gain is the tagger?** Part of it is rate budget: 10 Hz in 2018 vs 60, 115, 170 Hz in 2022, 2023, 2024
- **Are the H → 4ℓ intervals sign-sensitive?** Not really: the small asymmetry comes from charm-loop interference in ggH and H → gg; the constraint is essentially on κ_c^2

References

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- $t\bar{t}$H(H → $c\bar{c}$) + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- VH(H → $c\bar{c}$) with 2016 data (analysis strategy): CMS, JHEP 03 (2020) 131, arXiv:1912.01662 (HIG-18-031)
- Boosted H → $c\bar{c}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH(H → γγ): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H → WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2: CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- γH + H → 4ℓ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H → 4ℓ: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H → J/ψ γ: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- H → μμ: CMS, JHEP 01 (2021) 148, arXiv:2009.04363 (HIG-19-006)
- Jet flavour tagging in Run 3: CMS-DP-2025-081; DeepJet: E. Bols et al., JINST 15 (2020) P12012
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $t\bar{t}$+b/c cross sections: CMS-PAS-TOP-26-003
- ATLAS VBF + VH(H → $c\bar{c}$): ATLAS, arXiv:2511.21911
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- Higgs combination: CMS, Rep. Prog. Phys. 89 (2026) 087801, arXiv:2602.18611 (HIG-21-018)
- Spontaneous flavour violation: D. Egana-Ugrinovic, S. Homiller, P. Meade, PRL 123 (2019) 031802, arXiv:1811.00017; κ_c in SFV: A. S. Giannakopoulou, P. Meade, M. Valli, "How charming can the Higgs be?", JHEP 02 (2025) 067, arXiv:2410.05236
- SM Higgs branching ratios: LHC Higgs Cross Section WG, CERN Yellow Report 4, CERN-2017-002-M, arXiv:1610.07922
- HL-LHC Yellow Report: M. Cepeda et al., CERN Yellow Rep. Monogr. 7 (2019) 221, arXiv:1902.00134